12 综合案例

12.1 项目背景

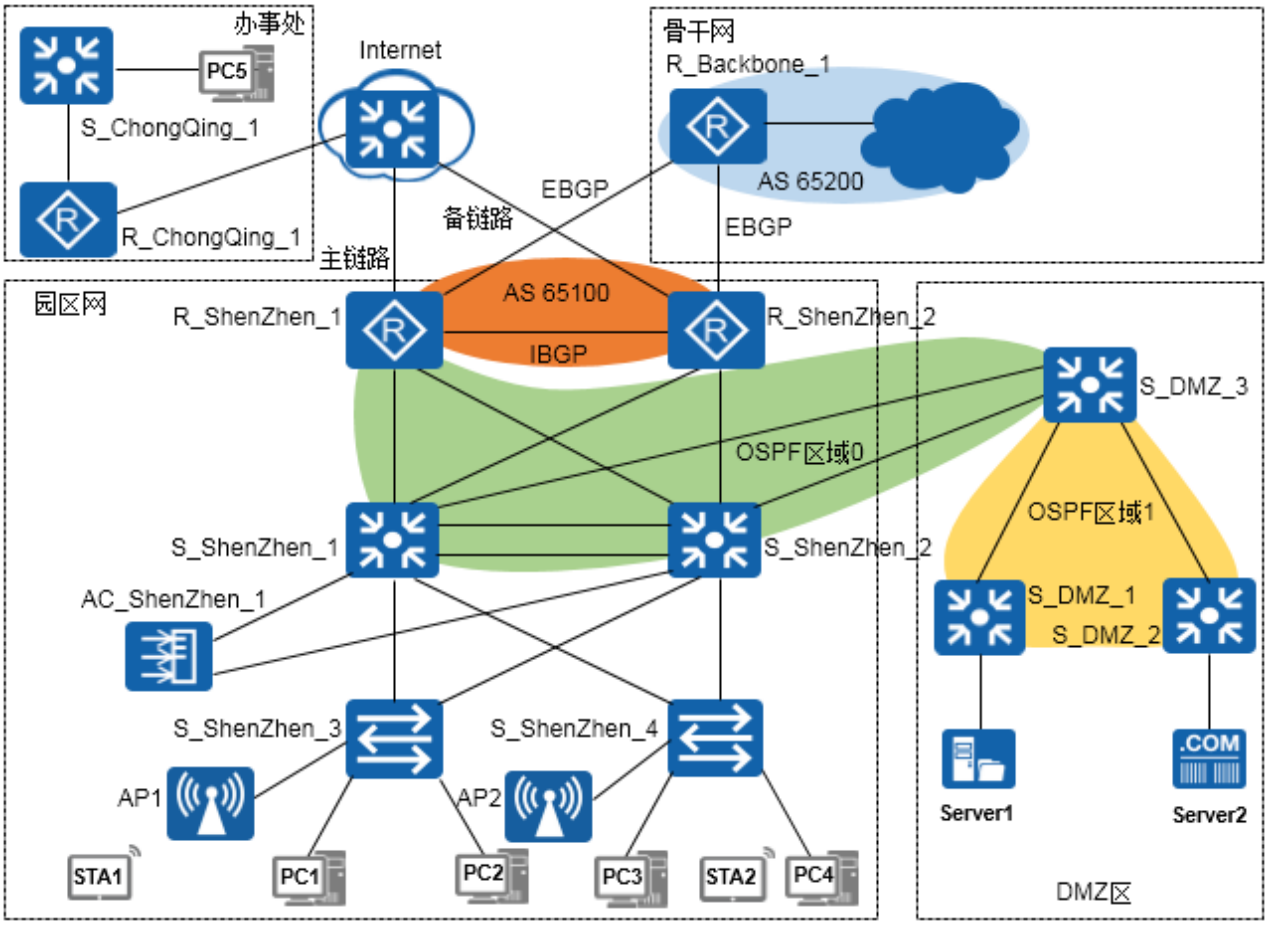
位于深圳的 A 公司有较大规模的园区网络，园区网络中有 DMZ 区，通过专线与企业骨干网相连；该公司在重庆设有办事处，办事处员工有访问公司 DMZ 区服务器的需求；公司租用两条线路接入 Internet。为了现实网络的优化、升级和未来网络的扩展，公司 IT 部门经理要求正在公司实习的五名同学利用两周时间熟悉目前企业网络架构和使用的核心技术，同时结合企业的发展战略设计与企业骨干网的连接并实施路由策略。五位同学利用一周时间在企业工程师的帮助下，基本熟悉了整个公司目前的网络架构。为了更好地适应未来工作岗位和做好技术人才储备，项目经理让五位同学利用实验室设备模拟整个网络的实现。同学们根据整个企业网络集成的预算并结合应用类型以及业务密集度的分析，得出网络数据负载、数据包流向、网络带宽、信息流特征、数据的重要程度、网络应用的安全性及可靠性和实时性等因素，从而确定网络总体需求框架。通过详细的需求分析，完整和量化地了解到网络系统的业务，并通过 eNSP 完成项目实施和测试。

12.2 项目目的

通过本项目可以掌握如下知识点和技能点，同时积累项目经验。

- 描述网络工程需求分析过程和方法。
- 掌握网络工程规划设计原则和步骤。
- 了解网络工程实施、网络工程验收和网络工程维护的过程。
- 掌握园区网络 VLAN、Trunk、链路聚合配置和调试。
- 掌握 MSTP 配置和调试。
- 掌握无线 AC 配置和调试。
- 掌握 DHCP 和 DHCP 安全配置和调试
- 掌握 VRRP、BFD 和 NAQ 配置和调试。
- 掌握 OSPF、IS-IS 和 BGP 配置和调试。
- 掌握路由引入和路由优化配置和调试。
- 掌握 NAT 配置和调试。
- 掌握 QoS 配置和调试。
- 掌握 IPSec VPN 配置和调试。
- 掌握 OSPFv3 配置和调试。

12.3 项目拓扑



12.4 项目规划

本项目的核心任务是完成路由协议、路由引入和路径控制、MSTP 部署、可靠性、服务质量、WLAN 和网络安全部署，为保持项目的完整性，需完成前期准备工作。

12.4.1 项目前期准备工作

步骤 1 VLAN 部署：在园区网、DMZ 区和 ISP 相应交换机上完成 VLAN 相关配置，包括 VLAN 创建和端口划分、Trunk 配置等。在两台核心交换机 S_ShenZhen_1 和 S_ShenZhen_2 之间采用手工模式配置链路聚合，增加核心交换机之间链路带宽和可靠性，实现流量负载分担，避免网络瓶颈。

步骤 2 IP 地址部署：在园区网、DMZ 区、骨干网和办事处设备上完成 IP 地址配置，包括配置路由器接口的 IP 地址、三层交换机创建 VLANIF 并配置 IP 地址以及配置计算机和服务器的 IP 地址、子网掩码和网关（DHCP 方式获得 IP 地址的主机除外）。

步骤 3 DHCP、交换机安全部署：将交换机 S_DMZ_1 配置成 DHCP 服务器，为园区网主机提供 DHCP 服务；S_ShenZhen_3、S_ShenZhen_4 配置为 DHCP 代理。用户的计算机 IP 必须动态

获取，不得手工设定，因此可以部署使用 DHCP Snooping 防止 DHCP 欺骗，然后在 DHCP Snooping 基础上部署 DAI 和 IPSG 等。当然不同 VLAN 的主机需要向 DHCP 服务器申请 IP 地址时，相关设备上还要配置 DHCP 中继。同时需要将无线控制器 AC_ShenZhen 配置为 DHCP 服务器。在接入交换机 S_ShenZhen_3 和 S_ShenZhen_4 上部署端口安全。

12.4.2 项目核心任务

步骤 1 MSTP 部署：在园区网交换机 S_ShenZhen_1、S_ShenZhen_2、S_ShenZhen_3、S_ShenZhen_4 和无线控制器 AC_ShenZhen 上配置 MSTP，MSTP 的域名为 ShenZhen，配置 MSTP 实例 1 和实例 2，实例 1 与 VLAN 的映射为 1-2，100；实例 2 与 VLAN 的映射为 3-4，200。控制 MSTP 各实例的根桥，以便实现负载均衡，提高设备的利用率和转发性能。S_ShenZhen_1 是实例 1 的根桥，优先级为 4096，是实例 2 的次根桥，优先级为 8192；S_ShenZhen_2 是实例 2 的根桥，优先级为 4096，是实例 1 的次根桥，优先级为 8192。将交换机 S_ShenZhen_3 和 S_ShenZhen_4 连接计算机的端口配置 STP 边缘端口实现 MSTP 的快速收敛。为了网络安全，建议在边缘端口上配置 BPDU 保护。

步骤 2 VRRP 部署：园区网两台核心层交换机 S_ShenZhen_1 和 S_ShenZhen_2 承担 VLAN 间路由功能，核心层交换机上 VLANIF 接口地址就是相关 VLAN 的计算机的默认网关，为了提高网关的稳定性和可靠性，实施 VRRP 技术实现网关的冗余。同时使用 BFD 技术提升 VRRP 的可靠性和快速收敛。控制不同 VLAN 的 VRRP 的主、备路由器，实现负载均衡，其中交换机 S_ShenZhen_1 是 1-2，100 的 Master。交换机 S_ShenZhen_2 是 3-4，200 的 Master。VRRP 的主、备路由器应该和 MSTP 的根桥配合，这样可以避免次优路径。为提高 VRRP 的切换速度，VRRP 和 BFD 技术进行联动。配置 VRRP 验证提高网络安全性。

步骤 3 OSPF 部署：园区网和 DMZ 区的路由协议选择 OSPF，OSPF 采用多区域设计，园区网属于区域 0，DMZ 区属于区域 1，区域 1 配置为完全末节区域。由两台核心交换机 S_ShenZhen_1 和 S_ShenZhen_2 承担主要的路由功能。所有运行 OSPF 的设备的参考带宽改为 1000M。区域 0 和区域 1 都采用 MD5 区域验证提高网络安全性。适当接口配置为静默接口提高网络安全性。在园区网边界路由器 R_ShenZhen_1 和 R_ShenZhen_2 上向 OSPF 网络注入默认路由，路由器 R_ShenZhen_1 通告默认路由的开销值为 10，路由器 R_ShenZhen_2 通告默认路由的开销值为 20，确保企业网络内的 OSPF 优先选择主链路达到 Internet。

步骤 4 WLAN 部署：选择使用 AC+瘦 AP 旁挂式组网方案在园区网中部署 WLAN，AP 是零配置，全部配置工作在有线网络和 AC 上进行，该方案既可以保证无缝漫游和可管理性，也方便后期网络扩展。无线控制器 AC_ShenZhen_1 通过双链路连接到核心交换机 S_ShenZhen_1 和 S_ShenZhen_2 的设计可以实现可靠性连接。两个 AP 接在靠近用户的接入层交换机 S_ShenZhen_3 和 S_ShenZhen_4 上。

步骤 5 BGP 部署：园区网与骨干网之间通过两条专线进行连接，并运行 BGP 路由协议。总部边界路由器 R_ShenZhen_1 和 R_ShenZhen_2 与骨干网边界路由器 R_Backbone_1 之间采用直连接口建立 EBGP 邻居，R_ShenZhen_1 和 R_ShenZhen_2 之间采用直连接口建立 IBGP 邻居。在路由器 R_ShenZhen_1 和 R_ShenZhen_2 做路由汇聚，使得路由器 R_Backbone_1 只学到 10.1.0.0/16 和 10.2.0.0/16 的聚合路由。通过 BGP 本地优先级属性控制 BGP 选路，在路由器 R_ShenZhen_1 和 R_ShenZhen_2 入向修改本地优先级属性，使得园区网去往骨干网 10.3.3.0/24 网络选择 R_ShenZhen_1 和 R_Backbone_1 之间链路，去往骨干网 10.3.4.0/24 网络选择 R_ShenZhen_2 和 R_Backbone_1 之间链路。通过 BGP MED 属性控制 BGP 选路，在路

由器 R_ShenZhen_1 和 R_ShenZhen_2 出向修改 MED 属性，控制路由器 R_Backbone_1 的 BGP 选路，使得骨干网访问园区网 10.1.0.0/16 网络选择 R_ShenZhen_1 和 R_Backbone_1 之间链路，访问 DMZ 区 10.2.0.0/16 网络选择 R_ShenZhen_2 和 R_Backbone_1 之间链路。

步骤 6 路由引入和路径控制部署：由于网络中存在 OSPF 和 BGP 路由协议，为了实现网络可达，需要部署路由引入，并通过路由策略进行路由优化和控制。在路由器 R_ShenZhen_1 和 R_ShenZhen_2 上均实施 BGP 和 OSPF 路由协议的双向路由引入。由于是双点双向路由引入，会带来次优路由、路由环路和路由反馈等问题，因此要实施路径控制。首先通过路由策略，对骨干网的路由在路由器 R_ShenZhen_1 和 R_ShenZhen_2 上修改优先级，只要低于 OSPF ASE 路由优先级 150 即可，本项目改为 140。为了避免路由环路，在路由引入时采用路由标记解决方案。

步骤 7 Internet 接入部署：为保证公司的用户连接到 Internet 不中断，园区网部署两条到 Internet 的线路，选择了不同营运商。由于不同运营商的线路质量和成本不一样，这两条线路采用主、备方式使用。园区网路由器 R_ShenZhen_1 使用主链路，路由器 R_ShenZhen_2 使用备链路。为保证主、备链路切换，采用静态路由与 NQA 联动，使用 NQA 监视主线路的状态，以便主链路异常时快速切换到备链路。

步骤 8 NAT 部署：为了减少向 ISP 购买公网 IP 地址的成本和公司内部 IP 编址的统一规划设计，园区网和办事处内部使用的是私有 IP 地址，因此需要在园区网到 Internet 的出口路由器 R_ShenZhen_1 和 R_ShenZhen_2 以及办事处路由器上部署 NAT，使得园区网用户和办事处用户可以访问 Internet 以及外网用户访问 DMZ 区主机。园区网有两个到 Internet 的出口，重庆办事处有一个到 Internet 的出口。因为重庆办事处需要建立 IPSec VPN 访问 DMZ 区服务器的资源，所以配置使用 NAT 的 ACL 要排除掉 IPSec VPN 的感兴趣流量。公司没有申请多余的公网 IP 地址，因为边界路由器各个出口已经有固定公网 IP 地址，所以使用 Easy IP 方式部署 NAT。公司内部有 Web 服务器 10.2.100.100 需要供外网访问，园区网两个出口均要做 NAT 服务器映射。

步骤 9 QoS 部署：为避免 Internet 链路视频、FTP 等流量影响公司的业务流量，需要部署 QoS 使得 Internet 链路优先保证关键应用，对 HTTP 流量和 IPSec VPN 的流量要保证带宽，HTTP 流量保证带宽为 1M，IPSec VPN 流量保证带宽为 512K。以上需求通过 MQC 的方法在两台边界路由器 R_ShenZhen_1 和 R_ShenZhen_2 上部署流量监管来实现。

步骤 10 IPSec VPN 部署：重庆办事处需要连接到 DMZ 区进行资源访问，由于专线价格的初装费和使用成本较高，因此采用 Site To Site IPSec VPN 技术作为解决方案。VPN 两端需要有固定公网 IP，会带来一定成本，但好处是 VPN 两端都可以主动建立 IPSec VPN 连接。企业总部有两个 Internet 出口，为了简化部署，本项目只在主链路连接的路由器 R_ShenZhen_1 上部署 IPSec VPN。建立 IPSec VPN 的感兴趣流量是重庆办事处的网络 10.4.0.0/16 到 DMZ 区的网络 10.2.0.0/16。

步骤 11 IPv6 部署：为了迎接 IPv6 时代的到来，园区网络和 DMZ 区先行部署 IPv6 网络用于测试和研究，暂时不考虑连接到 Internet。仅在园区网核心交换机上和 DMZ 区的网络设备上配置路由协议实现网络互通，选择 OSPFv3 作为路由协议。本项目只关注 IPv6 路由，暂时不考虑 IPv6 的 VRRP 问题。

交换机 VLAN 规划表、设备接口连接规划表和设备接口 IP 地址规划表如下。

表12-1 交换机 VLAN 规划表

设备	VLAN ID	IP 地址段	备注
S_ShenZhen_1	VLAN1	10.1.1.0/24	园区网
	VLAN2	10.1.2.0/24	园区网
	VLAN3	10.1.3.0/24	园区网
	VLAN4	10.1.4.0/24	园区网
	VLAN51	10.1.51.0/24	S_ShenZhen_1 和 R_ShenZhen_1 、R_ShenZhen_2 对接
	VLAN53	10.1.53.0/24	S_ShenZhen_1 和 S_DMZ_3 对接
	VLAN100	10.1.100.0/24	园区网，WLAN 管理 VLAN
	VLAN200	10.1.200.0/24	园区网，WLAN 业务 VLAN
S_ShenZhen_2	VLAN1	10.1.1.0/24	园区网
	VLAN2	10.1.2.0/24	园区网
	VLAN3	10.1.3.0/24	园区网
	VLAN4	10.1.4.0/24	园区网
	VLAN52	10.1.52.0/24	S_ShenZhen_2 和 R_ShenZhen_1 、R_ShenZhen_2 对接
	VLAN54	10.1.54.0/24	S_ShenZhen_2 和 S_DMZ_3 对接
	VLAN100	10.1.100.0/24	园区网，WLAN 管理 VLAN
	VLAN200	10.1.200.0/24	园区网，WLAN 业务 VLAN
S_ShenZhen_3	VLAN1	10.1.1.0/24	园区网
	VLAN2	10.1.2.0/24	园区网
	VLAN3	10.1.3.0/24	园区网
	VLAN4	10.1.4.0/24	园区网
	VLAN100	10.1.100.0/24	园区网，WLAN 管理 VLAN
	VLAN200	10.1.200.0/24	园区网，WLAN 业务 VLAN
S_ShenZhen_4	VLAN1	10.1.1.0/24	园区网
	VLAN2	10.1.2.0/24	园区网
	VLAN3	10.1.3.0/24	园区网
	VLAN4	10.1.4.0/24	园区网
	VLAN100	10.1.100.0/24	园区网，WLAN 管理 VLAN
	VLAN200	10.1.200.0/24	园区网，WLAN 业务 VLAN
S_DMZ_1	VLAN1	10.2.1.0/30	S_DMZ_1 和 S_DMZ_3 对接
	VLAN2	10.2.3.0/24	接 DMZ 区服务器
S_DMZ_2	VLAN1	10.2.2.0/30	S_DMZ_2 和 S_DMZ_3 对接
	VLAN2	10.2.4.0/24	接 DMZ 区服务器
S_DMZ_3	VLAN1	10.1.53.0/30	S_ShenZhen_1 和 S_DMZ_3 对接
	VLAN2	10.1.54.0/30	S_ShenZhen_2 和 S_DMZ_3 对接
	VLAN3	10.2.1.0/30	S_DMZ_1 和 S_DMZ_3 对接
	VLAN4	10.2.2.0/30	S_DMZ_2 和 S_DMZ_3 对接
ISP	VLAN1	61.1.1.0/30	和 R_ChongQing_1 对接
	VLAN2	202.96.1.0/30	和 R_ShenZhen_1 对接
	VLAN3	202.96.2.0/30	和 R_ShenZhen_2 对接

AC_ShenZhen_1	VLAN100	10.1.100.0/24	园区网，WLAN 管理 VLAN
	VLAN200	10.1.200.0/24	园区网，WLAN 业务 VLAN

表12-2 设备接口连接规划表

设备	接口	接口类型	VLAN	链路聚合	对端设备及接口
S_ShenZhen_1	G0/0/1	Trunk			S_ShenZhen_3 G0/0/1
	G0/0/2	Trunk			S_ShenZhen_4 G0/0/1
	G0/0/3	Access	VLAN 51		R_ShenZhen_1 G0/0/0
	G0/0/4	Access	VLAN 51		R_ShenZhen_2 G0/0/0
	G0/0/5	Trunk			AC_ShenZhen G0/0/1
	G0/0/6	Access	VLAN 53		S_DMZ_3 G0/0/1
	G0/0/23	Trunk		是	S_ShenZhen_2 G0/0/23
	G0/0/24	Trunk		是	S_ShenZhen_2 G0/0/24
S_ShenZhen_2	G0/0/1	Trunk			S_ShenZhen_3 G0/0/2
	G0/0/2	Trunk			S_ShenZhen_4 G0/0/2
	G0/0/3	Access	VLAN 52		R_ShenZhen_1 G0/0/1
	G0/0/4	Access	VLAN 52		R_ShenZhen_2 G0/0/1
	G0/0/5	Trunk			AC_ShenZhen G0/0/2
	G0/0/6	Access	VLAN 54		S_DMZ_3 G0/0/2
	G0/0/23	Trunk		是	S_ShenZhen_1 G0/0/23
	G0/0/24	Trunk		是	S_ShenZhen_1 G0/0/24
S_ShenZhen_3	E0/0/1	Access	VLAN 1		PC1
	E0/0/2	Access	VLAN 2		PC2
	E0/0/22	Trunk			AP1
	G0/0/1	Trunk			S_ShenZhen_1 G0/0/1
	G0/0/2	Trunk			S_ShenZhen_2 G0/0/1
S_ShenZhen_4	E0/0/1	Access	VLAN 3		PC3
	E0/0/2	Access	VLAN 4		PC4
	E0/0/22	Trunk			AP2
	G0/0/1	Trunk			S_ShenZhen_1 G0/0/2
	G0/0/2	Trunk			S_ShenZhen_2 G0/0/2
ISP	G0/0/1	Access	VLAN 1		R_ChongQing_1 G0/0/0
	G0/0/2	Access	VLAN 2		R_ShenZhen_1 G1/0/0
	G0/0/3	Access	VLAN 3		R_ShenZhen_2 G1/0/0
R_ShenZhen_1	G0/0/0				S_ShenZhen_1 G0/0/3
	G0/0/1				S_ShenZhen_2 G0/0/3
	G0/0/2				R_Backbone_1 G0/0/0
	G1/0/0				ISP G0/0/2
	G2/0/0				R_ShenZhen_2 G2/0/0
R_ShenZhen_2	G0/0/0				S_ShenZhen_1 G0/0/4
	G0/0/1				S_ShenZhen_2 G0/0/4
	G0/0/2				R_Backbone_1 G0/0/1

	G1/0/0				ISP G0/0/3
	G2/0/0				R_ShenZhen_1 G2/0/0
S_DMZ_3	G0/0/1	Access	VLAN 1		S_ShenZhen_1 G0/0/6
	G0/0/2	Access	VLAN 2		S_ShenZhen_2 G0/0/6
	G0/0/3	Access	VLAN 3		S_DMZ_1 G0/0/1
	G0/0/4	Access	VLAN 4		S_DMZ_2 G0/0/1
S_DMZ_1	G0/0/1	Access	VLAN 1		S_DMZ_3 G0/0/3
	G0/0/3	Access	VLAN 2		Server1
S_DMZ_2	G0/0/1	Access	VLAN 1		S_DMZ_3 G0/0/4
	G0/0/3	Access	VLAN 2		Server2
R_Backbone_1	G0/0/0				R_ShenZhen_1 G0/0/2
	G0/0/1				R_ShenZhen_2 G0/0/2
R_ChongQing_1	G0/0/0				ISP G0/0/1
	G0/0/1				S_ChongQing_1 G0/0/1

表12-3 设备接口 IPv4 地址规划表

设备	接口	IP 地址	备注
S_ShenZhen_1	VLANIF1	10.1.1.1/24	
	VLANIF2	10.1.2.1/24	
	VLANIF3	10.1.3.1/24	
	VLANIF4	10.1.4.1/24	
	VLANIF51	10.1.51.1/24	
	VLANIF53	10.1.53.1/30	
	VLANIF100	10.1.100.1/24	
	VLANIF200	10.1.200.1/24	
S_ShenZhen_2	VLANIF1	10.1.1.2/24	
	VLANIF2	10.1.2.2/24	
	VLANIF3	10.1.3.2/24	
	VLANIF4	10.1.4.2/24	
	VLANIF52	10.1.52.1/24	
	VLANIF54	10.1.54.1/30	
	VLANIF100	10.1.100.2/24	
	VLANIF200	10.1.200.2/24	
S_DMZ_3	VLANIF1	10.1.53.2/50	
	VLANIF2	10.1.54.2/30	
	VLANIF3	10.2.1.1/30	
	VLANIF4	10.2.2.1/30	
S_DMZ_2	VLANIF1	10.2.2.2/30	
	VLANIF2	10.2.4.254/24	
S_DMZ_1	VLANIF1	10.2.1.2/30	
	VLANIF2	10.2.3.254/24	
AC_ShenZhen	VLANIF100	10.1.100.3/24	

	VLANIF200	10.1.200.3/24	
R_ShenZhen_1	G0/0/0	10.1.51.2/24	
	G0/0/1	10.1.52.2/24	
	G0/0/2	172.16.1.1/20	
	G1/0/0	202.96.1.2/30	
	G2/0/0	10.1.55.1/30	
R_ShenZhen_2	G0/0/0	10.1.51.3/24	
	G0/0/1	10.1.52.3/24	
	G0/0/2	172.16.2.1/30	
	G1/0/0	202.96.2.2/30	
	G2/0/0	10.1.55.2/30	
ISP	VLANIF1	61.1.1.1/30	
	VLANIF2	202.96.1.1/30	
	VLANIF3	202.96.2.1/30	
	LoopBack1	8.8.8.8/24	
R_Backbone_1	G0/0/0	172.16.1.2/30	
	G0/0/1	172.16.2.2/30	
	LoopBack0	10.3.3.1/24	
	LoopBack1	10.3.4.1/24	
R_ChongQing_1	G0/0/0	61.1.1.2/30	
	G0/0/1	10.4.1.254/24	
PC1		DHCP	
PC2		DHCP	
PC3		DHCP	
PC4		DHCP	
PC5		10.4.1.100	网关：10.4.1.254
Server1		10.2.3.100	网关：10.2.3.254
Server2		10.2.4.100	网关：10.2.4.254
STA1		DHCP	
STA2		DHCP	

表12-4 设备接口 IPv6 地址规划表

VLAN 或者设备	接口	IP 地址	备注
S_ShenZhen_1	VLANIF1	2001:AAAA:BBBB:1::1/64	
	VLANIF2	2001:AAAA:BBBB:2::1/64	
	VLANIF3	2001:AAAA:BBBB:3::1/64	
	VLANIF4	2001:AAAA:BBBB:4::1/64	
	VLANIF53	2001:AAAA:BBBB:35::1/126	
S_ShenZhen_2	VLANIF1	2001:AAAA:BBBB:1::2/64	
	VLANIF2	2001:AAAA:BBBB:2::2/64	
	VLANIF3	2001:AAAA:BBBB:3::2/64	
	VLANIF4	2001:AAAA:BBBB:4::2/64	
	VLANIF54	2001:AAAA:BBBB:36::1/126	

S_DMZ_1	VLANIF1	2001:AAAA:BBBB:20::2/126	
	VLANIF2	2001:AAAA:BBBB:37::1/64	
S_DMZ_2	VLANIF1	2001:AAAA:BBBB:21::2/126	
	VLANIF2	2001:AAAA:BBBB:38::1/64	
S_DMZ_3	VLANIF1	2001:AAAA:BBBB:35::2/126	
	VLANIF2	2001:AAAA:BBBB:36::2/126	
	VLANIF3	2001:AAAA:BBBB:20::1/126	
	VLANIF4	2001:AAAA:BBBB:21::1/126	

12.5 项目实施

12.5.1 项目准备工作

步骤 1 配置 VLAN

在园区网、DMZ 区和 ISP 相应交换机上完成 VLAN 相关配置，包括 VLAN 创建和端口划分、Trunk 配置等。在两台核心交换机 S_ShenZhen_1 和 S_ShenZhen_2 之间采用手工模式配置链路聚合，增加核心交换机之间链路带宽和可靠性，实现流量负载分担，避免网络瓶颈。

- 配置交换机 S_ShenZhen_1

```
[S_ShenZhen_1]vlan batch 2 to 4 51 to 54 100 200
[S_ShenZhen_1]interface Eth-Trunk1
[S_ShenZhen_1-Eth-Trunk1]port link-type trunk
[S_ShenZhen_1-Eth-Trunk1]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_1-Eth-Trunk1]quit
[S_ShenZhen_1]interface GigabitEthernet0/0/1
[S_ShenZhen_1-GigabitEthernet0/0/1]port link-type trunk
[S_ShenZhen_1-GigabitEthernet0/0/1]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_1-GigabitEthernet0/0/1]quit
[S_ShenZhen_1]interface GigabitEthernet0/0/2
[S_ShenZhen_1-GigabitEthernet0/0/2]port link-type trunk
[S_ShenZhen_1-GigabitEthernet0/0/2]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_1-GigabitEthernet0/0/2]quit
[S_ShenZhen_1]interface GigabitEthernet0/0/3
[S_ShenZhen_1-GigabitEthernet0/0/3]port link-type access
[S_ShenZhen_1-GigabitEthernet0/0/3]port default vlan 51
[S_ShenZhen_1-GigabitEthernet0/0/3]quit
[S_ShenZhen_1]interface GigabitEthernet0/0/4
[S_ShenZhen_1-GigabitEthernet0/0/4]port link-type access
[S_ShenZhen_1-GigabitEthernet0/0/4]port default vlan 51
[S_ShenZhen_1-GigabitEthernet0/0/4]quit
[S_ShenZhen_1]interface GigabitEthernet0/0/5
[S_ShenZhen_1-GigabitEthernet0/0/5]port link-type trunk
[S_ShenZhen_1-GigabitEthernet0/0/5]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_1-GigabitEthernet0/0/5]quit
[S_ShenZhen_1]interface GigabitEthernet0/0/6
[S_ShenZhen_1-GigabitEthernet0/0/6]port link-type access
[S_ShenZhen_1-GigabitEthernet0/0/6]port default vlan 53
[S_ShenZhen_1-GigabitEthernet0/0/6]interface GigabitEthernet0/0/23
[S_ShenZhen_1-GigabitEthernet0/0/23]eth-trunk 1
[S_ShenZhen_1-GigabitEthernet0/0/23]quit
```

```
[S_ShenZhen_1]interface GigabitEthernet0/0/24
[S_ShenZhen_1-GigabitEthernet0/0/24]eth-trunk 1
```

- 配置交换机 S_ShenZhen_2

```
[S_ShenZhen_2]vlan batch 2 to 4 51 to 54 100 200
[S_ShenZhen_2]interface Eth-Trunk1
[S_ShenZhen_2-Eth-Trunk1]port link-type trunk
[S_ShenZhen_2-Eth-Trunk1]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_2-Eth-Trunk1]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/1
[S_ShenZhen_2-GigabitEthernet0/0/1]port link-type trunk
[S_ShenZhen_2-GigabitEthernet0/0/1]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_2-GigabitEthernet0/0/1]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/2
[S_ShenZhen_2-GigabitEthernet0/0/2]port link-type trunk
[S_ShenZhen_2-GigabitEthernet0/0/2]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_2-GigabitEthernet0/0/2]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/3
[S_ShenZhen_2-GigabitEthernet0/0/3]port link-type access
[S_ShenZhen_2-GigabitEthernet0/0/3]port default vlan 52
[S_ShenZhen_2-GigabitEthernet0/0/3]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/4
[S_ShenZhen_2-GigabitEthernet0/0/4]port link-type access
[S_ShenZhen_2-GigabitEthernet0/0/4]port default vlan 52
[S_ShenZhen_2-GigabitEthernet0/0/4]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/5
[S_ShenZhen_2-GigabitEthernet0/0/5]port link-type trunk
[S_ShenZhen_2-GigabitEthernet0/0/5]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_2-GigabitEthernet0/0/5]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/6
[S_ShenZhen_2-GigabitEthernet0/0/6]port link-type access
[S_ShenZhen_2-GigabitEthernet0/0/6]port default vlan 54
[S_ShenZhen_2-GigabitEthernet0/0/6]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/23
[S_ShenZhen_2-GigabitEthernet0/0/23]eth-trunk 1
[S_ShenZhen_2-GigabitEthernet0/0/23]quit
[S_ShenZhen_2]interface GigabitEthernet0/0/24
[S_ShenZhen_2-GigabitEthernet0/0/24]eth-trunk 1
```

- 配置交换机 S_ShenZhen_3

```
[S_ShenZhen_3]vlan batch 2 to 4 100 200
[S_ShenZhen_3]interface Ethernet0/0/1
[S_ShenZhen_3-Ethernet0/0/1]port link-type access
[S_ShenZhen_3-Ethernet0/0/1]quit
[S_ShenZhen_3]interface Ethernet0/0/2
[S_ShenZhen_3-Ethernet0/0/2]port link-type access
[S_ShenZhen_3-Ethernet0/0/2]port default vlan 2
[S_ShenZhen_3-Ethernet0/0/2]quit
[S_ShenZhen_3]interface Ethernet0/0/22
[S_ShenZhen_3-Ethernet0/0/22]port link-type trunk
[S_ShenZhen_3-Ethernet0/0/22]port trunk pvid vlan 100
[S_ShenZhen_3-Ethernet0/0/22]port trunk allow-pass vlan 100 200
[S_ShenZhen_3-Ethernet0/0/22]quit
[S_ShenZhen_3]interface GigabitEthernet0/0/1
[S_ShenZhen_3-GigabitEthernet0/0/1]port link-type trunk
[S_ShenZhen_3-GigabitEthernet0/0/1]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_3-GigabitEthernet0/0/1]quit
[S_ShenZhen_3]interface GigabitEthernet0/0/2
```

```
[S_ShenZhen_3-GigabitEthernet0/0/2]port link-type trunk
[S_ShenZhen_3-GigabitEthernet0/0/2]port trunk allow-pass vlan 2 to 4094
```

- 配置交换机 S_ShenZhen_4

```
[S_ShenZhen_4]vlan batch 2 to 4 100 200
[S_ShenZhen_4]interface Ethernet0/0/1
[S_ShenZhen_4-Ethernet0/0/1]port link-type access
[S_ShenZhen_4-Ethernet0/0/1]port default vlan 3
[S_ShenZhen_4-Ethernet0/0/1]quit
[S_ShenZhen_4]interface Ethernet0/0/2
[S_ShenZhen_4-Ethernet0/0/2]port link-type access
[S_ShenZhen_4-Ethernet0/0/2]port default vlan 4
[S_ShenZhen_4-Ethernet0/0/2]quit
[S_ShenZhen_4]interface Ethernet0/0/22
[S_ShenZhen_4-Ethernet0/0/22]port link-type trunk
[S_ShenZhen_4-Ethernet0/0/22]port trunk pvid vlan 100
[S_ShenZhen_4-Ethernet0/0/22]port trunk allow-pass vlan 100 200
[S_ShenZhen_4-Ethernet0/0/22]quit
[S_ShenZhen_4]interface GigabitEthernet0/0/1
[S_ShenZhen_4-GigabitEthernet0/0/1]port link-type trunk
[S_ShenZhen_4-GigabitEthernet0/0/1]port trunk allow-pass vlan 2 to 4094
[S_ShenZhen_4-GigabitEthernet0/0/1]quit
[S_ShenZhen_4]interface GigabitEthernet0/0/2
[S_ShenZhen_4-GigabitEthernet0/0/2]port link-type trunk
[S_ShenZhen_4-GigabitEthernet0/0/2]port trunk allow-pass vlan 2 to 4094
```

- 配置交换机 S_DMZ_1

```
[S_DMZ_1]vlan batch 2
[S_DMZ_1]interface GigabitEthernet0/0/1
[S_DMZ_1-GigabitEthernet0/0/1]port link-type access
[S_DMZ_1-GigabitEthernet0/0/1]quit
[S_DMZ_1]interface GigabitEthernet0/0/3
[S_DMZ_1-GigabitEthernet0/0/3]port link-type access
[S_DMZ_1-GigabitEthernet0/0/3]port default vlan 2
```

- 配置交换机 S_DMZ_2

```
[S_DMZ_2]vlan batch 2
[S_DMZ_2]interface GigabitEthernet0/0/1
[S_DMZ_2-GigabitEthernet0/0/1]port link-type access
[S_DMZ_2-GigabitEthernet0/0/1]quit
[S_DMZ_2]interface GigabitEthernet0/0/3
[S_DMZ_2-GigabitEthernet0/0/3]port link-type access
[S_DMZ_2-GigabitEthernet0/0/3]port default vlan 2
```

- 配置交换机 S_DMZ_3

```
[S_DMZ_3]vlan batch 2 to 4
[S_DMZ_3]interface GigabitEthernet0/0/1
[S_DMZ_3-GigabitEthernet0/0/1]port link-type access
[S_DMZ_3-GigabitEthernet0/0/1]quit
[S_DMZ_3]interface GigabitEthernet0/0/2
[S_DMZ_3-GigabitEthernet0/0/2]port link-type access
[S_DMZ_3-GigabitEthernet0/0/2]port default vlan 2
[S_DMZ_3-GigabitEthernet0/0/2]quit
[S_DMZ_3]interface GigabitEthernet0/0/3
[S_DMZ_3-GigabitEthernet0/0/3]port link-type access
[S_DMZ_3-GigabitEthernet0/0/3]port default vlan 3
[S_DMZ_3-GigabitEthernet0/0/3]quit
[S_DMZ_3]interface GigabitEthernet0/0/4
```

```
[S_DMZ_3-GigabitEthernet0/0/4]port link-type access
[S_DMZ_3-GigabitEthernet0/0/4]port default vlan 4
[S_DMZ_3-GigabitEthernet0/0/4]quit
```

- 配置交换机 ISP

```
[ISP]vlan batch 2 to 3
[ISP]interface GigabitEthernet0/0/1
[ISP-GigabitEthernet0/0/1]port link-type access
[ISP-GigabitEthernet0/0/1]quit
[ISP]interface GigabitEthernet0/0/2
[ISP-GigabitEthernet0/0/2]port link-type access
[ISP-GigabitEthernet0/0/2]port default vlan 2
[ISP-GigabitEthernet0/0/2]quit
[ISP]interface GigabitEthernet0/0/3
[ISP-GigabitEthernet0/0/3]port link-type access
[ISP-GigabitEthernet0/0/3]port default vlan 3
```

- 配置控制器 AC_ShenZhen

```
[AC_ShenZhen]vlan batch 100 200
[AC_ShenZhen]interface GigabitEthernet0/0/1
[AC_ShenZhen-GigabitEthernet0/0/1]port link-type trunk
[AC_ShenZhen-GigabitEthernet0/0/1]port trunk allow-pass vlan 2 to 4094
[AC_ShenZhen-GigabitEthernet0/0/1]quit
[AC_ShenZhen]interface GigabitEthernet0/0/2
[AC_ShenZhen-GigabitEthernet0/0/2]port link-type trunk
[AC_ShenZhen-GigabitEthernet0/0/2]port trunk allow-pass vlan 2 to 4094
```

步骤 2 配置 IP 地址

在园区网、DMZ 区、骨干网和办事处设备上完成 IP 地址配置，包括配置路由器接口的 IP 地址、三层交换机创建 VLANIF 并配置 IP 地址以及配置计算机和服务器的 IP 地址、子网掩码和网关（DHCP 方式获得 IP 地址的主机除外）。

- 配置交换机 S_ShenZhen_1

```
[S_ShenZhen_1]interface Vlanif1
[S_ShenZhen_1-Vlanif1]ip address 10.1.1.1 255.255.255.0
[S_ShenZhen_1-Vlanif1]quit
[S_ShenZhen_1]interface Vlanif2
[S_ShenZhen_1-Vlanif2]ip address 10.1.2.1 255.255.255.0
[S_ShenZhen_1-Vlanif2]quit
[S_ShenZhen_1]interface Vlanif3
[S_ShenZhen_1-Vlanif3]ip address 10.1.3.1 255.255.255.0
[S_ShenZhen_1-Vlanif3]quit
[S_ShenZhen_1]interface Vlanif4
[S_ShenZhen_1-Vlanif4]ip address 10.1.4.1 255.255.255.0
[S_ShenZhen_1-Vlanif4]quit
[S_ShenZhen_1]interface Vlanif51
[S_ShenZhen_1-Vlanif51]ip address 10.1.51.1 255.255.255.0
[S_ShenZhen_1-Vlanif51]quit
[S_ShenZhen_1]interface Vlanif53
[S_ShenZhen_1-Vlanif53]ip address 10.1.53.1 255.255.255.252
[S_ShenZhen_1-Vlanif53]quit
[S_ShenZhen_1]interface Vlanif100
[S_ShenZhen_1-Vlanif100]ip address 10.1.100.1 255.255.255.0
[S_ShenZhen_1-Vlanif100]quit
[S_ShenZhen_1]interface Vlanif200
[S_ShenZhen_1-Vlanif200]ip address 10.1.200.1 255.255.255.0
```

- 配置交换机 S_ShenZhen_2

```
[S_ShenZhen_2]interface Vlanif1
[S_ShenZhen_2-Vlanif1]ip address 10.1.1.2 255.255.255.0
[S_ShenZhen_2-Vlanif1]quit
[S_ShenZhen_2]interface Vlanif2
[S_ShenZhen_2-Vlanif2]ip address 10.1.2.2 255.255.255.0
[S_ShenZhen_2-Vlanif2]quit
[S_ShenZhen_2]interface Vlanif3
[S_ShenZhen_2-Vlanif3]ip address 10.1.3.2 255.255.255.0
[S_ShenZhen_2-Vlanif3]quit
[S_ShenZhen_2]interface Vlanif4
[S_ShenZhen_2-Vlanif4]ip address 10.1.4.2 255.255.255.0
[S_ShenZhen_2-Vlanif4]quit
[S_ShenZhen_2]interface Vlanif52
[S_ShenZhen_2-Vlanif52]ip address 10.1.52.1 255.255.255.0
[S_ShenZhen_2-Vlanif52]quit
[S_ShenZhen_2]interface Vlanif54
[S_ShenZhen_2-Vlanif54]ip address 10.1.54.1 255.255.255.252
[S_ShenZhen_2-Vlanif54]quit
[S_ShenZhen_2]interface Vlanif100
[S_ShenZhen_2-Vlanif100]ip address 10.1.100.2 255.255.255.0
[S_ShenZhen_2-Vlanif100]quit
[S_ShenZhen_2]interface Vlanif200
[S_ShenZhen_2-Vlanif200]ip address 10.1.200.2 255.255.255.0
```

- 配置控制器 AC_ShenZhen

```
[AC_ShenZhen]interface Vlanif100
[AC_ShenZhen-Vlanif100]ip address 10.1.100.3 255.255.255.0
[AC_ShenZhen-Vlanif100]quit
[AC_ShenZhen]interface Vlanif200
[AC_ShenZhen-Vlanif200]ip address 10.1.200.3 255.255.255.0
[AC_ShenZhen-Vlanif200]quit
[AC_ShenZhen]ip route-static 0.0.0.0 0.0.0.0 10.1.100.254
```

- 配置交换机 S_DMZ_1

```
[S_DMZ_1]interface Vlanif1
[S_DMZ_1-Vlanif1]ip address 10.2.1.2 255.255.255.252
[S_DMZ_1-Vlanif1]quit
[S_DMZ_1]interface Vlanif2
[S_DMZ_1-Vlanif2]ip address 10.2.3.254 255.255.255.0
```

- 配置交换机 S_DMZ_2

```
[S_DMZ_2]interface Vlanif1
[S_DMZ_2-Vlanif1]ip address 10.2.2.2 255.255.255.252
[S_DMZ_2-Vlanif1]quit
[S_DMZ_2]interface Vlanif2
[S_DMZ_2-Vlanif2]ip address 10.2.4.254 255.255.255.0
```

- 配置交换机 S_DMZ_3

```
[S_DMZ_3]interface Vlanif1
[S_DMZ_3-Vlanif1]ip address 10.1.53.2 255.255.255.252
[S_DMZ_3-Vlanif1]quit
[S_DMZ_3]interface Vlanif2
[S_DMZ_3-Vlanif2]ip address 10.1.54.2 255.255.255.252
[S_DMZ_3-Vlanif2]quit
[S_DMZ_3]interface Vlanif3
[S_DMZ_3-Vlanif3]ip address 10.2.1.1 255.255.255.252
```

```
[S_DMZ_3-Vlanif3]quit
[S_DMZ_3]interface Vlanif4
[S_DMZ_3-Vlanif4]ip address 10.2.2.1 255.255.255.252
```

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]interface GigabitEthernet0/0/0
[R_ShenZhen_1-GigabitEthernet0/0/0]ip address 10.1.51.2 255.255.255.0
[R_ShenZhen_1-GigabitEthernet0/0/0]quit
[R_ShenZhen_1]interface GigabitEthernet0/0/1
[R_ShenZhen_1-GigabitEthernet0/0/1]ip address 10.1.52.2 255.255.255.0
[R_ShenZhen_1-GigabitEthernet0/0/1]quit
[R_ShenZhen_1]interface GigabitEthernet0/0/2
[R_ShenZhen_1-GigabitEthernet0/0/2]ip address 172.16.1.1 255.255.255.252
[R_ShenZhen_1-GigabitEthernet0/0/2]quit
[R_ShenZhen_1]interface GigabitEthernet1/0/0
[R_ShenZhen_1-GigabitEthernet1/0/0]ip address 202.96.1.2 255.255.255.252
[R_ShenZhen_1-GigabitEthernet1/0/0]quit
[R_ShenZhen_1]interface GigabitEthernet2/0/0
[R_ShenZhen_1-GigabitEthernet2/0/0]ip address 10.1.55.1 255.255.255.252
```

- 配置路由器 R_ShenZhen_2

```
[R_ShenZhen_2]interface GigabitEthernet0/0/0
[R_ShenZhen_2-GigabitEthernet0/0/0]ip address 10.1.51.3 255.255.255.0
[R_ShenZhen_2-GigabitEthernet0/0/0]quit
[R_ShenZhen_2]interface GigabitEthernet0/0/1
[R_ShenZhen_2-GigabitEthernet0/0/1]ip address 10.1.52.3 255.255.255.0
[R_ShenZhen_2-GigabitEthernet0/0/1]quit
[R_ShenZhen_2]interface GigabitEthernet0/0/2
[R_ShenZhen_2-GigabitEthernet0/0/2]ip address 172.16.2.1 255.255.255.252
[R_ShenZhen_2-GigabitEthernet0/0/2]quit
[R_ShenZhen_2]interface GigabitEthernet1/0/0
[R_ShenZhen_2-GigabitEthernet1/0/0]ip address 202.96.2.2 255.255.255.252
[R_ShenZhen_2-GigabitEthernet1/0/0]quit
[R_ShenZhen_2]interface GigabitEthernet2/0/0
[R_ShenZhen_2-GigabitEthernet2/0/0]ip address 10.1.55.2 255.255.255.252
```

- 配置交换机 ISP

```
[ISP]interface Vlanif1
[ISP-Vlanif1]ip address 61.1.1.1 255.255.255.252
[ISP-Vlanif1]quit
[ISP]interface Vlanif2
[ISP-Vlanif2]ip address 202.96.1.1 255.255.255.252
[ISP-Vlanif2]quit
[ISP]interface Vlanif3
[ISP-Vlanif3]ip address 202.96.2.1 255.255.255.252
[ISP-Vlanif3]quit
[ISP]interface LoopBack1
[ISP-LoopBack1]ip address 8.8.8.8 255.255.255.0
```

- 配置路由器 R_Backbone_1

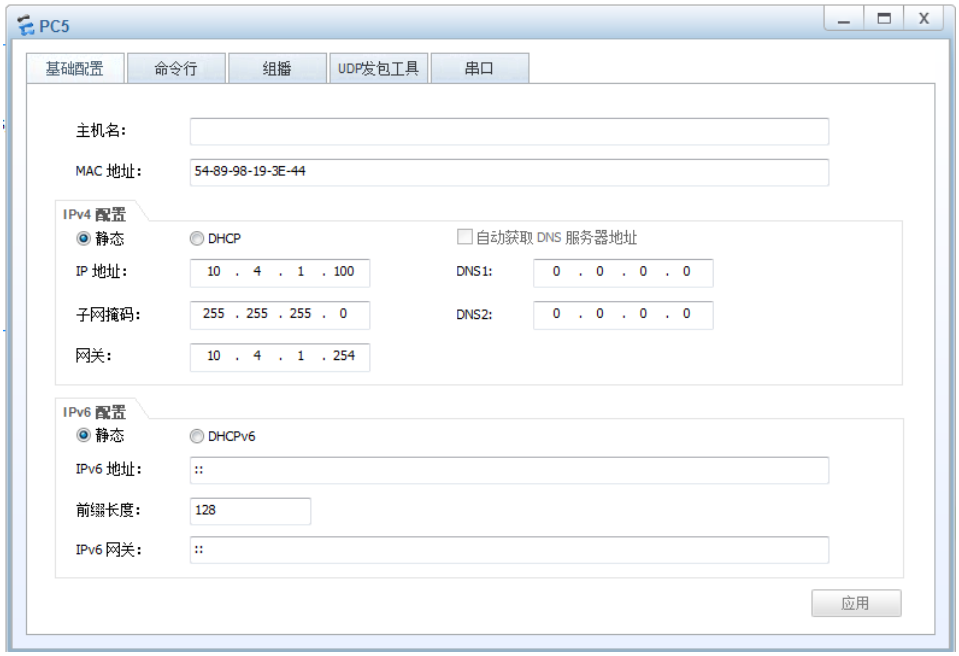
```
[R_Backbone_1]interface GigabitEthernet0/0/0
[R_Backbone_1-GigabitEthernet0/0/0]ip address 172.16.1.2 255.255.255.252
[R_Backbone_1-GigabitEthernet0/0/0]quit
[R_Backbone_1]interface GigabitEthernet0/0/1
[R_Backbone_1-GigabitEthernet0/0/1]ip address 172.16.2.2 255.255.255.252
[R_Backbone_1-GigabitEthernet0/0/1]quit
[R_Backbone_1]interface LoopBack0
[R_Backbone_1-LoopBack0]ip address 10.3.3.1 255.255.255.0
```

```
[R_Backbone_1-LoopBack0]quit
[R_Backbone_1]interface LoopBack1
[R_Backbone_1-LoopBack1]ip address 10.3.4.1 255.255.255.0
```

- 配置路由器 R_ChongQing_1

```
[R_ChongQing_1]interface GigabitEthernet0/0/0
[R_ChongQing_1-GigabitEthernet0/0/0]ip address 61.1.1.2 255.255.255.252
[R_ChongQing_1-GigabitEthernet0/0/0]quit
[R_ChongQing_1]interface GigabitEthernet0/0/1
[R_ChongQing_1-GigabitEthernet0/0/1]ip address 10.4.1.254 255.255.255.0
```

- 以主机 PC5 和服务器 Sever1 为例配置主机和服务器 IP 地址、掩码和网关。



步骤 3 配置 DHCP、交换机安全

将交换机 S_DMZ_1 配置成 DHCP 服务器，为园区网主机提供 DHCP 服务；S_ShenZhen_3、S_ShenZhen_4 配置为 DHCP 代理。用户的计算机 IP 必须动态获取，不得手工设定，因此可以部署使用 DHCP Snooping 防止 DHCP 欺骗，然后在 DHCP Snooping 基础上部署 DAI 和 IPSG 等。当然不同 VLAN 的主机需要向 DHCP 服务器申请 IP 地址时，相关设备上还要配置 DHCP 中继。同时需要将无线控制器 AC_ShenZhen 配置为 DHCP 服务器。在接入交换机 S_ShenZhen_3 和 S_ShenZhen_4 上部署端口安全。

- 配置交换机 S_DMZ_1

```
[S_DMZ_1]dhcp enable
[S_DMZ_1]ip pool vlan01
[S_DMZ_1-ip-pool-vlan01]gateway-list 10.1.1.254
[S_DMZ_1-ip-pool-vlan01]network 10.1.1.0 mask 255.255.255.0
[S_DMZ_1-ip-pool-vlan01]excluded-ip-address 10.1.1.1 10.1.1.10
[S_DMZ_1-ip-pool-vlan01]dns-list 10.2.100.100
[S_DMZ_1-ip-pool-vlan01]quit
[S_DMZ_1]ip pool vlan02
[S_DMZ_1-ip-pool-vlan02]gateway-list 10.1.2.254
[S_DMZ_1-ip-pool-vlan02]network 10.1.2.0 mask 255.255.255.0
[S_DMZ_1-ip-pool-vlan02]excluded-ip-address 10.1.2.1 10.1.2.10
[S_DMZ_1-ip-pool-vlan02]dns-list 10.2.100.100
[S_DMZ_1-ip-pool-vlan02]quit
[S_DMZ_1]ip pool vlan03
[S_DMZ_1-ip-pool-vlan03]gateway-list 10.1.3.254
[S_DMZ_1-ip-pool-vlan03]network 10.1.3.0 mask 255.255.255.0
[S_DMZ_1-ip-pool-vlan03]excluded-ip-address 10.1.3.1 10.1.3.10
[S_DMZ_1-ip-pool-vlan03]dns-list 10.2.100.100
[S_DMZ_1-ip-pool-vlan03]quit
[S_DMZ_1]ip pool vlan04
[S_DMZ_1-ip-pool-vlan04]gateway-list 10.1.4.254
[S_DMZ_1-ip-pool-vlan04]network 10.1.4.0 mask 255.255.255.0
[S_DMZ_1-ip-pool-vlan04]excluded-ip-address 10.1.4.1 10.1.4.10
[S_DMZ_1-ip-pool-vlan04]dns-list 10.2.100.100
[S_DMZ_1-ip-pool-vlan04]quit
[S_DMZ_1]interface Vlanif1
[S_DMZ_1-Vlanif1]dhcp select global
```

- 配置交换机 S_ShenZhen_1

```
[S_ShenZhen_1]dhcp enable
[S_ShenZhen_1]interface Vlanif1
[S_ShenZhen_1-Vlanif1]dhcp select relay
[S_ShenZhen_1-Vlanif1]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_1-Vlanif1]quit
[S_ShenZhen_1]interface Vlanif2
[S_ShenZhen_1-Vlanif2]dhcp select relay
[S_ShenZhen_1-Vlanif2]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_1-Vlanif2]quit
[S_ShenZhen_1]interface Vlanif3
[S_ShenZhen_1-Vlanif3]dhcp select relay
[S_ShenZhen_1-Vlanif3]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_1-Vlanif3]quit
[S_ShenZhen_1]interface Vlanif4
[S_ShenZhen_1-Vlanif4]dhcp select relay
[S_ShenZhen_1-Vlanif4]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_1-Vlanif4]quit
```

- 配置交换机 S_ShenZhen_2

```
[S_ShenZhen_2]dhcp enable
[S_ShenZhen_2]interface Vlanif1
[S_ShenZhen_2-Vlanif1]dhcp select relay
[S_ShenZhen_2-Vlanif1]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_2-Vlanif1]quit
[S_ShenZhen_2]interface Vlanif2
[S_ShenZhen_2-Vlanif2]dhcp select relay
[S_ShenZhen_2-Vlanif2]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_2-Vlanif2]quit
[S_ShenZhen_2]interface Vlanif3
[S_ShenZhen_2-Vlanif3]dhcp select relay
[S_ShenZhen_2-Vlanif3]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_2-Vlanif3]quit
[S_ShenZhen_2]interface Vlanif4
[S_ShenZhen_2-Vlanif4]dhcp select relay
[S_ShenZhen_2-Vlanif4]dhcp relay server-ip 10.2.1.2
[S_ShenZhen_2-Vlanif4]quit
```

- 配置交换机 S_ShenZhen_3

```
[S_ShenZhen_3]dhcp enable
[S_ShenZhen_3]dhcp snooping enable
[S_ShenZhen_3]vlan 1
[S_ShenZhen_3-vlan1]dhcp snooping enable
[S_ShenZhen_3-vlan1]quit
[S_ShenZhen_3]vlan 2
[S_ShenZhen_3-vlan2]dhcp snooping enable
[S_ShenZhen_3-vlan2]quit
[S_ShenZhen_3]vlan 3
[S_ShenZhen_3-vlan3]dhcp snooping enable
[S_ShenZhen_3-vlan3]quit
[S_ShenZhen_3]vlan 4
[S_ShenZhen_3-vlan4]dhcp snooping enable
[S_ShenZhen_3-vlan4]quit
[S_ShenZhen_3]interface GigabitEthernet0/0/1
[S_ShenZhen_3-GigabitEthernet0/0/1]dhcp snooping enable
[S_ShenZhen_3-GigabitEthernet0/0/1]dhcp snooping trusted
[S_ShenZhen_3-GigabitEthernet0/0/1]quit
[S_ShenZhen_3]interface GigabitEthernet0/0/2
[S_ShenZhen_3-GigabitEthernet0/0/2]dhcp snooping enable
[S_ShenZhen_3-GigabitEthernet0/0/2]dhcp snooping trusted
[S_ShenZhen_3-GigabitEthernet0/0/2]quit
[S_ShenZhen_3]interface Ethernet0/0/1
[S_ShenZhen_3-Ethernet0/0/1]port-security enable
[S_ShenZhen_3-Ethernet0/0/1]arp anti-attack check user-bind enable
[S_ShenZhen_3-Ethernet0/0/1]ip source check user-bind enable
[S_ShenZhen_3-Ethernet0/0/1]quit
[S_ShenZhen_3]interface Ethernet0/0/2
[S_ShenZhen_3-Ethernet0/0/2]port-security enable
[S_ShenZhen_3-Ethernet0/0/2]arp anti-attack check user-bind enable
[S_ShenZhen_3-Ethernet0/0/2]ip source check user-bind enable
[S_ShenZhen_3-Ethernet0/0/2]quit
```

- 配置交换机 S_ShenZhen_4

```
[S_ShenZhen_4]dhcp enable
[S_ShenZhen_4]dhcp snooping enable
[S_ShenZhen_4]vlan 1
[S_ShenZhen_4-vlan1]dhcp snooping enable
[S_ShenZhen_4-vlan1]quit
```

```
[S_ShenZhen_4]vlan 2
[S_ShenZhen_4-vlan2]dhcp snooping enable
[S_ShenZhen_4-vlan2]quit
[S_ShenZhen_4]vlan 3
[S_ShenZhen_4-vlan3]dhcp snooping enable
[S_ShenZhen_4-vlan3]quit
[S_ShenZhen_4]vlan 4
[S_ShenZhen_4-vlan4]dhcp snooping enable
[S_ShenZhen_4-vlan4]quit
[S_ShenZhen_4]interface GigabitEthernet0/0/1
[S_ShenZhen_4-GigabitEthernet0/0/1]dhcp snooping enable
[S_ShenZhen_4-GigabitEthernet0/0/1]dhcp snooping trusted
[S_ShenZhen_4-GigabitEthernet0/0/1]quit
[S_ShenZhen_4]interface GigabitEthernet0/0/2
[S_ShenZhen_4-GigabitEthernet0/0/2]dhcp snooping enable
[S_ShenZhen_4-GigabitEthernet0/0/2]dhcp snooping trusted
[S_ShenZhen_4-GigabitEthernet0/0/2]quit
[S_ShenZhen_4]interface Ethernet0/0/1
[S_ShenZhen_4-Ethernet0/0/1]port-security enable
[S_ShenZhen_4-Ethernet0/0/1]arp anti-attack check user-bind enable
[S_ShenZhen_4-Ethernet0/0/1]ip source check user-bind enable
[S_ShenZhen_4-Ethernet0/0/1]quit
[S_ShenZhen_4]interface Ethernet0/0/2
[S_ShenZhen_4-Ethernet0/0/2]port-security enable
[S_ShenZhen_4-Ethernet0/0/2]arp anti-attack check user-bind enable
[S_ShenZhen_4-Ethernet0/0/2]ip source check user-bind enable
[S_ShenZhen_4-Ethernet0/0/2]quit
```

- 配置无线控制器 AC_ShenZhen

```
[AC_ShenZhen]dhcp enable
[AC_ShenZhen]interface Vlanif100
[AC_ShenZhen-Vlanif100]dhcp select interface
[AC_ShenZhen-Vlanif100]dhcp server gateway-list 10.1.100.254
[AC_ShenZhen-Vlanif100]dhcp server dns-list 10.2.100.100
[AC_ShenZhen-Vlanif100]quit
[AC_ShenZhen]interface Vlanif200
[AC_ShenZhen-Vlanif200]dhcp select interface
[AC_ShenZhen-Vlanif200]dhcp server gateway-list 10.1.200.254
[AC_ShenZhen-Vlanif200]dhcp server dns-list 10.2.100.100
```

验证以上配置。

```
[S_DMZ_1]display dhcp statistics
Input: total 13 packets, discarded 0 packets
  Bootp request          :          0,  Bootp reply          :          0
  Discover               :          9,  Offer                :          0
  Request                :          4,  Ack                  :          0
  Release                :          0,  Nak                  :          0
  Decline                :          0,  Inform               :          0

Output: total 12 packets, discarded 4 packets

[S_ShenZhen_2]display dhcp relay all
 DHCP relay agent running information of interface Vlanif1 :
 Server IP address [01] : 10.2.1.2
 Gateway address in use : 10.1.1.254

 DHCP relay agent running information of interface Vlanif2 :
 Server IP address [01] : 10.2.1.2
```

```
 Gateway address in use : 10.1.2.254

 DHCP relay agent running information of interface Vlanif3 :
 Server IP address [01] : 10.2.1.2
 Gateway address in use : 10.1.3.254

 DHCP relay agent running information of interface Vlanif4 :
 Server IP address [01] : 10.2.1.2
 Gateway address in use : 10.1.4.254

[S_ShenZhen_4]display dhcp snooping vlan 3
DHCP snooping running information for VLAN 3 :
 DHCP snooping                                  : Enable
 Dhcp user max number                           : 1024        (default)
 Current dhcp user number                       : 1
 Check dhcp-giaddr                              : Disable    (default)
 Check dhcp-chaddr                              : Disable    (default)
 Check dhcp-request                             : Disable    (default)
 Check dhcp-rate                                : Disable    (default)
```

12.5.2 项目核心任务

步骤 1 配置 MSTP

在园区网交换机 S_ShenZhen_1、S_ShenZhen_2、S_ShenZhen_3、S_ShenZhen_4 和无线控制器 AC_ShenZhen 上配置 MSTP，MSTP 的域名为 ShenZhen，配置 MSTP 实例 1 和实例 2，实例 1 与 VLAN 的映射为 1-2，100；实例 2 与 VLAN 的映射为 3-4，200。控制 MSTP 各实例的根桥，以便实现负载均衡，提高设备的利用率和转发性能。S_ShenZhen_1 是实例 1 的根桥，优先级为 4096，是实例 2 的次根桥，优先级为 8192；S_ShenZhen_2 是实例 2 的根桥，优先级为 4096，是实例 1 的次根桥，优先级为 8192。将交换机 S_ShenZhen_3 和 S_ShenZhen_4 连接计算机的端口配置 STP 边缘端口实现 MSTP 的快速收敛。为了网络安全，建议在边缘端口上配置 BPDU 保护。

- 配置交换机 S_ShenZhen_1

```
[S_ShenZhen_1]stp instance 1 priority 4096
[S_ShenZhen_1]stp instance 2 priority 8192
[S_ShenZhen_1]stp region-configuration
[S_ShenZhen_1-mst-region]region-name ShenZhen
[S_ShenZhen_1-mst-region]instance 1 vlan 1 to 2    100
[S_ShenZhen_1-mst-region]instance 2 vlan 3 to 4    200
[S_ShenZhen_1-mst-region]active region-configuration
[S_ShenZhen_1-mst-region]quit
```

- 配置交换机 S_ShenZhen_2

```
[S_ShenZhen_2]stp instance 1 priority 8192
[S_ShenZhen_2]stp instance 2 priority 4096
[S_ShenZhen_2]stp region-configuration
[S_ShenZhen_2-mst-region]region-name ShenZhen
[S_ShenZhen_2-mst-region]instance 1 vlan 1 to 2    100
[S_ShenZhen_2-mst-region]instance 2 vlan 3 to 4    200
[S_ShenZhen_2-mst-region]active region-configuration
[S_ShenZhen_2-mst-region]quit
```

- 配置交换机 S_ShenZhen_3

```
[S_ShenZhen_3]stp region-configuration
```

```
[S_ShenZhen_3-mst-region]region-name ShenZhen
[S_ShenZhen_3-mst-region]instance 1 vlan 1 to 2    100
[S_ShenZhen_3-mst-region]instance 2 vlan 3 to 4    200
[S_ShenZhen_3-mst-region]active region-configuration
[S_ShenZhen_3-mst-region]quit
[S_ShenZhen_3]stp bpdu-protection
[S_ShenZhen_3]interface Ethernet0/0/1
[S_ShenZhen_3-Ethernet0/0/1]stp edged-port enable
[S_ShenZhen_3-Ethernet0/0/1]quit
[S_ShenZhen_3]interface Ethernet0/0/2
[S_ShenZhen_3-Ethernet0/0/2]stp edged-port enable
```

- 配置交换机 S_ShenZhen_4

```
[S_ShenZhen_4]stp region-configuration
[S_ShenZhen_4-mst-region]region-name ShenZhen
[S_ShenZhen_4-mst-region]instance 1 vlan 1 to 2    100
[S_ShenZhen_4-mst-region]instance 2 vlan 3 to 4    200
[S_ShenZhen_4-mst-region]active region-configuration
[S_ShenZhen_4-mst-region]quit
[S_ShenZhen_4]stp bpdu-protection
[S_ShenZhen_4]interface Ethernet0/0/1
[S_ShenZhen_4-Ethernet0/0/1]stp edged-port enable
[S_ShenZhen_4-Ethernet0/0/1]quit
[S_ShenZhen_4]interface Ethernet0/0/2
[S_ShenZhen_4-Ethernet0/0/2]stp edged-port enable
```

- 配置无线控制器 AC_ShenZhen

```
[AC_ShenZhen]stp enable
[AC_ShenZhen]stp region-configuration
[AC_ShenZhen-mst-region]region-name ShenZhen
[AC_ShenZhen-mst-region]instance 1 vlan 1 to 2    100
[AC_ShenZhen-mst-region]instance 2 vlan 3 to 4    200
[AC_ShenZhen-mst-region]active region-configuration
[AC_ShenZhen-mst-region]quit
```

验证以上配置。

```
[S_ShenZhen_1]display stp brief
 MSTID    Port                          Role    STP State         Protection
    0       GigabitEthernet0/0/1          DESI    FORWARDING          NONE
    0       GigabitEthernet0/0/2          DESI    FORWARDING          NONE
    0       GigabitEthernet0/0/3          DESI    FORWARDING          NONE
    0       GigabitEthernet0/0/4          DESI    FORWARDING          NONE
    0       GigabitEthernet0/0/5          ROOT    FORWARDING          NONE
    0       GigabitEthernet0/0/6          DESI    FORWARDING          NONE
    0       Eth-Trunk1                    DESI    FORWARDING          NONE
    1       GigabitEthernet0/0/1          DESI    FORWARDING          NONE
    1       GigabitEthernet0/0/2          DESI    FORWARDING          NONE
    1       GigabitEthernet0/0/5          DESI    FORWARDING          NONE
    1       Eth-Trunk1                    DESI    FORWARDING          NONE
    2       GigabitEthernet0/0/1          DESI    FORWARDING          NONE
    2       GigabitEthernet0/0/2          DESI    FORWARDING          NONE
    2       GigabitEthernet0/0/5          DESI    FORWARDING          NONE
    2       Eth-Trunk1                    ROOT    FORWARDING          NONE

[S_ShenZhen_2]display stp brief
 MSTID    Port                          Role     STP State         Protection
    0       GigabitEthernet0/0/1          DESI    FORWARDING          NONE
    0       GigabitEthernet0/0/2          DESI    FORWARDING          NONE
```

```
    0      GigabitEthernet0/0/3          DESI   FORWARDING        NONE
    0      GigabitEthernet0/0/4          DESI   FORWARDING        NONE
    0      GigabitEthernet0/0/5          ROOT   FORWARDING        NONE
    0      GigabitEthernet0/0/6          DESI   FORWARDING        NONE
    0      Eth-Trunk1                    ALTE   DISCARDING        NONE
    1      GigabitEthernet0/0/1          DESI   FORWARDING        NONE
    1      GigabitEthernet0/0/2          DESI   FORWARDING        NONE
    1      GigabitEthernet0/0/5          DESI   FORWARDING        NONE
    1      Eth-Trunk1                    ROOT   FORWARDING        NONE
    2      GigabitEthernet0/0/1          DESI   FORWARDING        NONE
    2      GigabitEthernet0/0/2          DESI   FORWARDING        NONE
    2      GigabitEthernet0/0/5          DESI   FORWARDING        NONE
    2      Eth-Trunk1                    DESI   FORWARDING        NONE

[S_ShenZhen_3]display stp brief
 MSTID    Port                          Role    STP State       Protection
    0      Ethernet0/0/1                 DESI   FORWARDING        BPDU
    0      Ethernet0/0/2                 DESI   FORWARDING        BPDU
    0      Ethernet0/0/22                DESI   FORWARDING        NONE
    0      GigabitEthernet0/0/1          ROOT   FORWARDING        NONE
    0      GigabitEthernet0/0/2          ALTE   DISCARDING        NONE
    1      Ethernet0/0/1                 DESI   FORWARDING        BPDU
    1      Ethernet0/0/2                 DESI   FORWARDING        BPDU
    1      Ethernet0/0/22                DESI   FORWARDING        NONE
    1      GigabitEthernet0/0/1          ROOT   FORWARDING        NONE
    1      GigabitEthernet0/0/2          ALTE   DISCARDING        NONE
    2      Ethernet0/0/22                DESI   FORWARDING        NONE
    2      GigabitEthernet0/0/1          ALTE   DISCARDING        NONE
    2      GigabitEthernet0/0/2          ROOT   FORWARDING        NONE


[S_ShenZhen_4]display stp brief
 MSTID    Port                          Role    STP State       Protection
    0      Ethernet0/0/1                 DESI   FORWARDING        BPDU
    0      Ethernet0/0/2                 DESI   FORWARDING        BPDU
    0      Ethernet0/0/22                DESI   FORWARDING        NONE
    0      GigabitEthernet0/0/1          ROOT   FORWARDING        NONE
    0      GigabitEthernet0/0/2          ALTE   DISCARDING        NONE
    1      Ethernet0/0/22                DESI   FORWARDING        NONE
    1      GigabitEthernet0/0/1          ROOT   FORWARDING        NONE
    1      GigabitEthernet0/0/2          ALTE   DISCARDING        NONE
    2      Ethernet0/0/1                 DESI   FORWARDING        BPDU
    2      Ethernet0/0/2                 DESI   FORWARDING        BPDU
    2      Ethernet0/0/22                DESI   FORWARDING        NONE
    2      GigabitEthernet0/0/1          ALTE   DISCARDING        NONE
    2      GigabitEthernet0/0/2          ROOT   FORWARDING        NONE

[AC_ShenZhen]display stp brief
 MSTID     Port                         Role    STP State       Protection
      0     GigabitEthernet0/0/1        DESI   FORWARDING        NONE
      0     GigabitEthernet0/0/2        DESI   FORWARDING        NONE
      1     GigabitEthernet0/0/1        ROOT   FORWARDING        NONE
      1     GigabitEthernet0/0/2        ALTE   DISCARDING        NONE
      2     GigabitEthernet0/0/1        ALTE   DISCARDING        NONE
      2     GigabitEthernet0/0/2        ROOT   FORWARDING        NONE
```

步骤 2 配置 VRRP

园区网两台核心层交换机 S_ShenZhen_1 和 S_ShenZhen_2 承担 VLAN 间路由功能，核心层交换机上 VLANIF 接口地址就是相关 VLAN 的计算机的默认网关，为了提高网关的稳定性和可靠性，实施 VRRP 技术实现网关的冗余。同时使用 BFD 技术提升 VRRP 的可靠性和快速收敛。控制不同 VLAN 的 VRRP 的主、备路由器，实现负载均衡，其中交换机 S_ShenZhen_1 是 1-2，100 的 Master。交换机 S_ShenZhen_2 是 3-4，200 的 Master。VRRP 的主、备路由器应该和 MSTP 的根桥配合，这样可以避免次优路径。为提高 VRRP 的切换速度，VRRP 和 BFD 技术进行联动。配置 VRRP 验证提高网络安全性。

- 配置交换机 S_ShenZhen_1

```
[S_ShenZhen_1]bfd
[S_ShenZhen_1-bfd]quit
[S_ShenZhen_1]bfd vlan03 bind peer-ip 10.1.3.2 source-ip 10.1.3.1 auto
[S_ShenZhen_1-bfd-session-vlan03]commit
[S_ShenZhen_1-bfd-session-vlan03]quit
[S_ShenZhen_1]bfd vlan04 bind peer-ip 10.1.4.2 source-ip 10.1.4.1 auto
[S_ShenZhen_1-bfd-session-vlan04]commit
[S_ShenZhen_1-bfd-session-vlan04]quit
[S_ShenZhen_1]interface Vlanif1
[S_ShenZhen_1-Vlanif1]vrrp vrid 1 virtual-ip 10.1.1.254
[S_ShenZhen_1-Vlanif1]vrrp vrid 1 priority 120
[S_ShenZhen_1-Vlanif1]vrrp vrid 1 preempt-mode timer delay 5
[S_ShenZhen_1-Vlanif1]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_1-Vlanif1]quit
[S_ShenZhen_1]interface Vlanif2
[S_ShenZhen_1-Vlanif2]vrrp vrid 1 virtual-ip 10.1.2.254
[S_ShenZhen_1-Vlanif2]vrrp vrid 1 priority 120
[S_ShenZhen_1-Vlanif2]vrrp vrid 1 preempt-mode timer delay 5
[S_ShenZhen_1-Vlanif2]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_1-Vlanif2]quit
[S_ShenZhen_1]interface Vlanif3
[S_ShenZhen_1-Vlanif3]vrrp vrid 1 virtual-ip 10.1.3.254
[S_ShenZhen_1-Vlanif3]vrrp vrid 1 track bfd-session session-name vlan03 increased 25
[S_ShenZhen_1-Vlanif3]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_1-Vlanif3]quit
[S_ShenZhen_1]interface Vlanif4
[S_ShenZhen_1-Vlanif4]vrrp vrid 1 virtual-ip 10.1.4.254
[S_ShenZhen_1-Vlanif4]vrrp vrid 1 track bfd-session session-name vlan04 increased 25
[S_ShenZhen_1-Vlanif4]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_1-Vlanif4]quit
[S_ShenZhen_1]interface Vlanif100
[S_ShenZhen_1-Vlanif100]vrrp vrid 1 virtual-ip 10.1.100.254
[S_ShenZhen_1-Vlanif100]vrrp vrid 1 priority 120
[S_ShenZhen_1-Vlanif100]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_1-Vlanif100]quit
[S_ShenZhen_1]interface Vlanif200
[S_ShenZhen_1-Vlanif200]vrrp vrid 1 virtual-ip 10.1.200.254
[S_ShenZhen_1-Vlanif200]vrrp vrid 1 authentication-mode md5 123456
```

- 配置交换机 S_ShenZhen_2

```
[S_ShenZhen_2]bfd
[S_ShenZhen_2-bfd]quit
[S_ShenZhen_2]bfd vlan01 bind peer-ip 10.1.1.1 source-ip 10.1.1.2 auto
[S_ShenZhen_2-bfd-session-vlan01]commit
[S_ShenZhen_2-bfd-session-vlan01]quit
[S_ShenZhen_2]bfd vlan02 bind peer-ip 10.1.2.1 source-ip 10.1.2.2 auto
[S_ShenZhen_2-bfd-session-vlan02]commit
```

```
[S_ShenZhen_2-bfd-session-vlan02]quit
[S_ShenZhen_2]interface Vlanif1
[S_ShenZhen_2-Vlanif1]vrrp vrid 1 virtual-ip 10.1.1.254
[S_ShenZhen_2-Vlanif1]vrrp vrid 1 track bfd-session session-name vlan01 increased 25
[S_ShenZhen_2-Vlanif1]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_2-Vlanif1]quit
[S_ShenZhen_2]interface Vlanif2
[S_ShenZhen_2-Vlanif2]vrrp vrid 1 virtual-ip 10.1.2.254
[S_ShenZhen_2-Vlanif2]vrrp vrid 1 track bfd-session session-name vlan02 increased 25
[S_ShenZhen_2-Vlanif2]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_2-Vlanif2]quit
[S_ShenZhen_2]interface Vlanif3
[S_ShenZhen_2-Vlanif3]vrrp vrid 1 virtual-ip 10.1.3.254
[S_ShenZhen_2-Vlanif3]vrrp vrid 1 priority 120
[S_ShenZhen_2-Vlanif3]vrrp vrid 1 preempt-mode timer delay 5
[S_ShenZhen_2-Vlanif3]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_2-Vlanif3]quit
[S_ShenZhen_2]interface Vlanif4
[S_ShenZhen_2-Vlanif4]vrrp vrid 1 virtual-ip 10.1.4.254
[S_ShenZhen_2-Vlanif4]vrrp vrid 1 priority 120
[S_ShenZhen_2-Vlanif4]vrrp vrid 1 preempt-mode timer delay 5
[S_ShenZhen_2-Vlanif4]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_2-Vlanif4]quit
[S_ShenZhen_2]interface Vlanif100
[S_ShenZhen_2-Vlanif100]vrrp vrid 1 virtual-ip 10.1.100.254
[S_ShenZhen_2-Vlanif100]vrrp vrid 1 authentication-mode md5 123456
[S_ShenZhen_2-Vlanif100]quit
[S_ShenZhen_2]interface Vlanif200
[S_ShenZhen_2-Vlanif200]vrrp vrid 1 virtual-ip 10.1.200.254
[S_ShenZhen_2-Vlanif200]vrrp vrid 1 priority 120
[S_ShenZhen_2-Vlanif200]vrrp vrid 1 authentication-mode md5 123456
```

验证以上配置。

```
[S_ShenZhen_1]display vrrp brief
VRID    State          Interface                    Type        Virtual IP
------------------------------------------------------------------------------
1       Master         Vlanif1                      Normal      10.1.1.254
1       Master         Vlanif2                      Normal      10.1.2.254
1       Backup         Vlanif3                      Normal      10.1.3.254
1       Backup         Vlanif4                      Normal      10.1.4.254
1       Master         Vlanif100                    Normal      10.1.100.254
1       Backup         Vlanif200                    Normal      10.1.200.254
------------------------------------------------------------------------------
Total:6      Master:3      Backup:3      Non-active:0

[S_ShenZhen_1]display bfd session all
--------------------------------------------------------------------------------
Local Remote      PeerIpAddr      State      Type           InterfaceName
--------------------------------------------------------------------------------

8204  8204        10.1.3.2        Up         S_AUTO_PEER        -
8205  8205        10.1.4.2        Up         S_AUTO_PEER        -
--------------------------------------------------------------------------------
        Total UP/DOWN Session Number : 2/0

[S_ShenZhen_2]display vrrp  brief
VRID    State          Interface                    Type        Virtual IP
------------------------------------------------------------------------------
```

```
1        Backup        Vlanif1                    Normal     10.1.1.254
1        Backup        Vlanif2                    Normal     10.1.2.254
1        Master        Vlanif3                    Normal     10.1.3.254
1        Master        Vlanif4                    Normal     10.1.4.254
1        Backup        Vlanif100                  Normal     10.1.100.254
1        Master        Vlanif200                  Normal     10.1.200.254
-------------------------------------------------------------
Total:6        Master:3        Backup:3        Non-active:0

[S_ShenZhen_2]display bfd session all
--------------------------------------------------------------------------------
Local Remote        PeerIpAddr        State        Type            InterfaceName
--------------------------------------------------------------------------------

8202   8202         10.1.1.1          Up           S_AUTO_PEER          -
8203   8203         10.1.2.1          Up           S_AUTO_PEER          -

--------------------------------------------------------------------------------
        Total UP/DOWN Session Number : 2/0
```

步骤 3 配置 OSPF

园区网和 DMZ 区的路由协议选择 OSPF，OSPF 采用多区域设计，园区网属于区域 0，DMZ 区属于区域 1，区域 1 配置为完全末节区域。由两台核心交换机 S_ShenZhen_1 和 S_ShenZhen_2 承担主要的路由功能。所有运行 OSPF 的设备的参考带宽改为 1000M。区域 0 和区域 1 都采用 MD5 区域验证提高网络安全性。适当接口配置为静默接口提高网络安全性。在园区网边界路由器 R_ShenZhen_1 和 R_ShenZhen_2 上向 OSPF 网络注入默认路由，路由器 R_ShenZhen_1 通告默认路由的开销值为 10，路由器 R_ShenZhen_2 通告默认路由的开销值为 20，确保企业网络内的 OSPF 优先选择主链路达到 Internet。

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]ospf 1 router-id 1.1.1.1
[R_ShenZhen_1-ospf-1]default-route-advertise cost 10 type 1
[R_ShenZhen_1-ospf-1]area 0.0.0.0
[R_ShenZhen_1-ospf-1-area-0.0.0.0]authentication-mode md5 1 cipher 123456
[R_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.51.2 0.0.0.0
[R_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.52.2 0.0.0.0
[R_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.55.1 0.0.0.0
[R_ShenZhen_1-ospf-1-area-0.0.0.0]quit
[R_ShenZhen_1-ospf-1]quit
[R_ShenZhen_1]interface GigabitEthernet0/0/0
[R_ShenZhen_1-GigabitEthernet0/0/0]ospf cost 2
[R_ShenZhen_1-GigabitEthernet0/0/0]ospf authentication-mode md5 1 cipher 123456
[R_ShenZhen_1-GigabitEthernet0/0/0]quit
[R_ShenZhen_1]interface GigabitEthernet0/0/1
[R_ShenZhen_1-GigabitEthernet0/0/1]ospf cost 2
[R_ShenZhen_1-GigabitEthernet0/0/1]ospf authentication-mode md5 1 cipher 123456
[R_ShenZhen_1-GigabitEthernet0/0/1]quit
[R_ShenZhen_1]interface GigabitEthernet2/0/0
[R_ShenZhen_1-GigabitEthernet2/0/0]ospf authentication-mode md5 1 cipher 123456
```

- 配置路由器 R_ShenZhen_2

```
[R_ShenZhen_2]ospf 1 router-id 2.2.2.2
[R_ShenZhen_2-ospf-1]default-route-advertise cost 20 type 1
[R_ShenZhen_2-ospf-1]area 0.0.0.0
```

```
[R_ShenZhen_2-ospf-1-area-0.0.0.0]authentication-mode md5 1 cipher 123456
[R_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.51.3 0.0.0.0
[R_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.52.3 0.0.0.0
[R_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.55.2 0.0.0.0
[R_ShenZhen_2-ospf-1-area-0.0.0.0]quit
[R_ShenZhen_2-ospf-1]quit
[R_ShenZhen_2]interface GigabitEthernet0/0/0
[R_ShenZhen_2-GigabitEthernet0/0/0]ospf cost 2
[R_ShenZhen_2-GigabitEthernet0/0/0]ospf authentication-mode md5 1 cipher 123456
[R_ShenZhen_2-GigabitEthernet0/0/0]quit
[R_ShenZhen_2]interface GigabitEthernet0/0/1
[R_ShenZhen_2-GigabitEthernet0/0/1]ospf cost 2
[R_ShenZhen_2-GigabitEthernet0/0/1]ospf authentication-mode md5 1 cipher 123456
[R_ShenZhen_2-GigabitEthernet0/0/1]quit
[R_ShenZhen_2]interface GigabitEthernet2/0/0
[R_ShenZhen_2-GigabitEthernet2/0/0]ospf authentication-mode md5 1 cipher 123456
```

- 配置交换机 S_ShenZhen_1

```
[S_ShenZhen_1]ospf 1 router-id 3.3.3.3
[S_ShenZhen_1-ospf-1]silent-interface Vlanif3
[S_ShenZhen_1-ospf-1]silent-interface Vlanif4
[S_ShenZhen_1-ospf-1]silent-interface Vlanif1
[S_ShenZhen_1-ospf-1]silent-interface Vlanif2
[S_ShenZhen_1-ospf-1]silent-interface Vlanif100
[S_ShenZhen_1-ospf-1]silent-interface Vlanif200
[S_ShenZhen_1-ospf-1]area 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]authentication-mode md5 1 cipher 123456
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.1.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.2.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.3.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.4.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.51.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.53.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.100.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]network 10.1.200.1 0.0.0.0
[S_ShenZhen_1-ospf-1-area-0.0.0.0]quit
[S_ShenZhen_1-ospf-1]quit
[S_ShenZhen_1]interface Vlanif51
[S_ShenZhen_1-Vlanif51]ospf authentication-mode md5 1 cipher 123456
[S_ShenZhen_1-Vlanif51]quit
[S_ShenZhen_1]interface Vlanif53
[S_ShenZhen_1-Vlanif53]ospf authentication-mode md5 1 cipher 123456
```

- 配置交换机 S_ShenZhen_2

```
[S_ShenZhen_2]ospf 1 router-id 4.4.4.4
[S_ShenZhen_2-ospf-1]silent-interface Vlanif3
[S_ShenZhen_2-ospf-1]silent-interface Vlanif4
[S_ShenZhen_2-ospf-1]silent-interface Vlanif1
[S_ShenZhen_2-ospf-1]silent-interface Vlanif2
[S_ShenZhen_2-ospf-1]silent-interface Vlanif100
[S_ShenZhen_2-ospf-1]silent-interface Vlanif200
[S_ShenZhen_2-ospf-1]area 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]authentication-mode md5 1 cipher 123456
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.1.2 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.2.2 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.3.2 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.4.2 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.54.1 0.0.0.0
```

```
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.52.1 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.100.1 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]network 10.1.200.2 0.0.0.0
[S_ShenZhen_2-ospf-1-area-0.0.0.0]quit
[S_ShenZhen_2-ospf-1]quit
[S_ShenZhen_2]interface Vlanif52
[S_ShenZhen_2-Vlanif52]ospf authentication-mode md5 1 cipher 123456
[S_ShenZhen_2-Vlanif52]quit
[S_ShenZhen_2]interface Vlanif54
[S_ShenZhen_2-Vlanif54]ospf authentication-mode md5 1 cipher 123456
```

- 配置交换机 S_DMZ_1

```
[S_DMZ_1]ospf 1 router-id 6.6.6.6
[S_DMZ_1-ospf-1]silent-interface Vlanif2
[S_DMZ_1-ospf-1]area 0.0.0.1
[S_DMZ_1-ospf-1-area-0.0.0.1]authentication-mode md5 1 cipher 123456
[S_DMZ_1-ospf-1-area-0.0.0.1]network 10.2.1.2 0.0.0.0
[S_DMZ_1-ospf-1-area-0.0.0.1]network 10.2.3.254 0.0.0.0
[S_DMZ_1-ospf-1-area-0.0.0.1]stub
[S_DMZ_1-ospf-1-area-0.0.0.1]quit
[S_DMZ_1-ospf-1]quit
[S_DMZ_1]interface Vlanif1
[S_DMZ_1-Vlanif1]ospf authentication-mode md5 1 cipher 123456
[S_DMZ_1-Vlanif1]quit
[S_DMZ_1]interface Vlanif2
[S_DMZ_1-Vlanif2]ospf authentication-mode md5 1 cipher 123456
```

- 配置交换机 S_DMZ_2

```
[S_DMZ_2]ospf 1 router-id 7.7.7.7
[S_DMZ_2-ospf-1]silent-interface Vlanif2
[S_DMZ_2-ospf-1]area 0.0.0.1
[S_DMZ_2-ospf-1-area-0.0.0.1]authentication-mode md5 1 cipher 123456
[S_DMZ_2-ospf-1-area-0.0.0.1]network 10.2.4.254 0.0.0.0
[S_DMZ_2-ospf-1-area-0.0.0.1]network 10.2.2.2 0.0.0.0
[S_DMZ_2-ospf-1-area-0.0.0.1]stub
[S_DMZ_2-ospf-1-area-0.0.0.1]quit
[S_DMZ_2-ospf-1]quit
[S_DMZ_2]interface Vlanif1
[S_DMZ_2-Vlanif1]ospf authentication-mode md5 1 cipher 123456
[S_DMZ_2-Vlanif1]quit
[S_DMZ_2]interface Vlanif2
[S_DMZ_2-Vlanif2]ospf authentication-mode md5 1 cipher 123456
```

- 配置交换机 S_DMZ_3

```
[S_DMZ_3]ospf 1 router-id 5.5.5.5
[S_DMZ_3-ospf-1]area 0.0.0.0
[S_DMZ_3-ospf-1-area-0.0.0.0]authentication-mode md5 1 cipher 123456
[S_DMZ_3-ospf-1-area-0.0.0.0]network 10.1.53.2 0.0.0.0
[S_DMZ_3-ospf-1-area-0.0.0.0]network 10.1.54.2 0.0.0.0
[S_DMZ_3-ospf-1-area-0.0.0.0]quit
[S_DMZ_3-ospf-1]area 0.0.0.1
[S_DMZ_3-ospf-1-area-0.0.0.1]abr-summary 10.2.0.0 255.255.0.0
[S_DMZ_3-ospf-1-area-0.0.0.1]authentication-mode md5 1 cipher 123456
[S_DMZ_3-ospf-1-area-0.0.0.1]network 10.2.1.1 0.0.0.0
[S_DMZ_3-ospf-1-area-0.0.0.1]network 10.2.2.1 0.0.0.0
[S_DMZ_3-ospf-1-area-0.0.0.1]stub no-summary
[S_DMZ_3-ospf-1-area-0.0.0.1]quit
[S_DMZ_3-ospf-1]quit
```

```
[S_DMZ_3]interface Vlanif1
[S_DMZ_3-Vlanif1]ospf authentication-mode md5 1 123456
[S_DMZ_3-Vlanif1]quit
[S_DMZ_3]interface Vlanif2
[S_DMZ_3-Vlanif2]ospf authentication-mode md5 1 123456
[S_DMZ_3-Vlanif2]quit
[S_DMZ_3]interface Vlanif3
[S_DMZ_3-Vlanif3]ospf authentication-mode md5 1 123456
[S_DMZ_3-Vlanif3]quit
[S_DMZ_3]interface Vlanif4
[S_DMZ_3-Vlanif4]ospf authentication-mode md5 1 123456
[S_DMZ_3-Vlanif4]quit
```

步骤 4 配置 WLAN

选择使用 AC+瘦 AP 旁挂式组网方案在园区网中部署 WLAN，AP 是零配置，全部配置工作在有线网络和 AC 上进行，该方案既可以保证无缝漫游和可管理性，也方便后期网络扩展。无线控制器 AC_ShenZhen_1 通过双链路连接到核心交换机 S_ShenZhen_1 和 S_ShenZhen_2 的设计可以实现可靠性连接。两个 AP 接在靠近用户的接入层交换机 S_ShenZhen_3 和 S_ShenZhen_4 上。

- 配置 AP 上线

```
[AC_ShenZhen]wlan
[AC_ShenZhen-wlan-view]ap-group name ap-group1
[AC_ShenZhen-wlan-ap-group-ap-group1]quit
[AC_ShenZhen-wlan-view]regulatory-domain-profile name default
[AC_ShenZhen-wlan-regulate-domain-default]country-code cn
[AC_ShenZhen-wlan-regulate-domain-default]quit
[AC_ShenZhen-wlan-view]ap-group name ap-group1
[AC_ShenZhen-wlan-ap-group-ap-group1]regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurations of the radio and reset
the AP. Continue?[Y/N]:y
[AC_ShenZhen-wlan-ap-group-ap-group1]quit
[AC_ShenZhen-wlan-view]quit
[AC_ShenZhen]capwap source interface vlanif 100
[AC_ShenZhen-wlan-view]ap auth-mode mac-auth
[AC_ShenZhen-wlan-view]ap-id 0 ap-mac ac85-3d92-3340
[AC_ShenZhen-wlan-ap-0]ap-name area_1
[AC_ShenZhen-wlan-ap-0]ap-group ap-group1
[AC_ShenZhen-wlan-ap-0]quit
[AC_ShenZhen-wlan-view]ap-id 1 ap-mac ac85-3d92-1b60
[AC_ShenZhen-wlan-ap-1]ap-name area_2
[AC_ShenZhen-wlan-ap-1]ap-group ap-group1
[AC_ShenZhen-wlan-ap-1]quit
```

验证以上配置，确认 AP 上线。

```
[AC_ShenZhen-wlan-view]display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor    : normal                [2]
-------------------------------------------------------------------------------
ID     MAC                     Name      Group        IP              Type                 State STA Uptime
-------------------------------------------------------------------------------
0      ac85-3d92-3340 area_1 ap-group1 10.1.100.207 AP6050DN            nor     1     36M:47S
1      ac85-3d92-1b60 area_2 ap-group1 10.1.100.237 AP6050DN            nor     1     36M:46S
```

```
--------------------------------------------------------------------------------
```

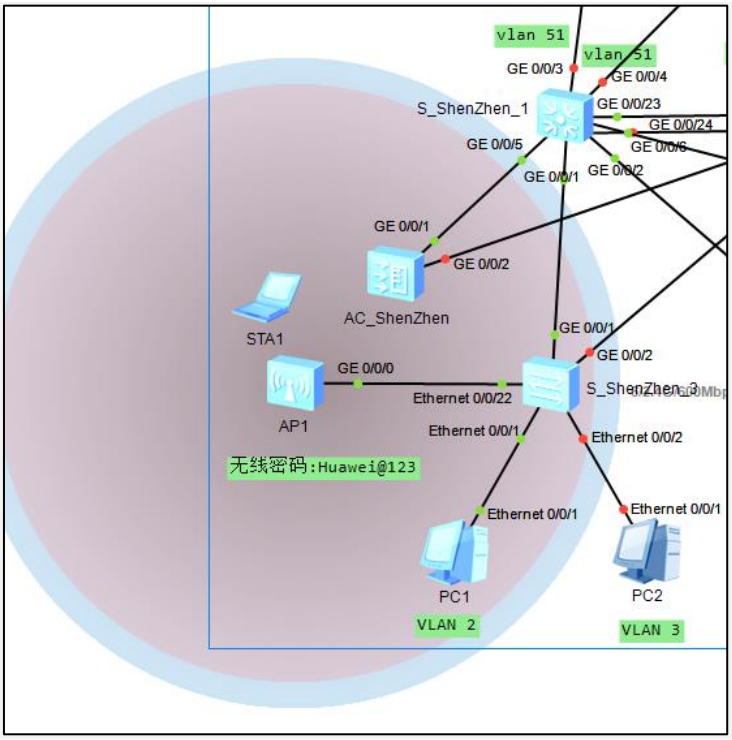
- 配置 WLAN 业务参数、AP 无线参数

```
[AC_ShenZhen-wlan-view]security-profile name wlan-net
[AC_ShenZhen-wlan-sec-prof-wlan-net]security wpa-wpa2 psk pass-phrase Huawei@123 aes
Warning: This action may cause service interruption. Continue?[Y/N]y
Info: This operation may take a few seconds, please wait.done.
[AC_ShenZhen-wlan-sec-prof-wlan-net]quit
[AC_ShenZhen-wlan-view]ssid-profile name wlan-net
[AC_ShenZhen-wlan-ssid-prof-wlan-net]ssid wlan-ne
[AC_ShenZhen-wlan-ssid-prof-wlan-net]quit
[AC_ShenZhen-wlan-view]vap-profile name wlan-net
[AC_ShenZhen-wlan-vap-prof-wlan-net]service-vlan vlan-id 200
[AC_ShenZhen-wlan-vap-prof-wlan-net]ssid-profile wlan-net
Warning: This action may cause service interruption. Continue?[Y/N]y

[AC_ShenZhen-wlan-vap-prof-wlan-net]security-profile wlan-net
[AC_ShenZhen-wlan-vap-prof-wlan-net]quit
[AC_ShenZhen-wlan-view]ap-group name ap-group1
[AC_ShenZhen-wlan-ap-group-ap-group1]radio 0
[AC_ShenZhen-wlan-group-radio-ap-group1/0]vap-profile wlan-net wlan 1
[AC_ShenZhen-wlan-group-radio-ap-group1/0]quit
[AC_ShenZhen-wlan-ap-group-ap-group1]radio 1
[AC_ShenZhen-wlan-group-radio-ap-group1/1]vap-profile wlan-net wlan 1
[AC_ShenZhen-wlan-group-radio-ap-group1/1]quit
[AC_ShenZhen-wlan-view]ap-id 0
[AC_ShenZhen-wlan-ap-0]radio 0
[AC_ShenZhen-wlan-radio-0/0]channel 20mhz 6
Warning: This action may cause service interruption. Continue?[Y/N]y
[AC_ShenZhen-wlan-radio-0/0]eirp 127
[AC_ShenZhen-wlan-radio-0/0]quit
[AC_ShenZhen-wlan-ap-0]radio 1
[AC_ShenZhen-wlan-radio-0/1]channel 20mhz 149
[AC_ShenZhen-wlan-radio-0/1]eirp 127
[AC_ShenZhen-wlan-radio-0/1]quit
[AC_ShenZhen-wlan-view]ap-id 1
[AC_ShenZhen-wlan-ap-1]radio 0
[AC_ShenZhen-wlan-radio-0/0]channel 20mhz 11
Warning: This action may cause service interruption. Continue?[Y/N]y
[AC_ShenZhen-wlan-radio-0/0]eirp 127
[AC_ShenZhen-wlan-radio-0/0]quit
[AC_ShenZhen-wlan-ap-1]radio 1
[AC_ShenZhen-wlan-radio-0/1]channel 20mhz 153
[AC_ShenZhen-wlan-radio-0/1]eirp 127
[AC_ShenZhen-wlan-radio-0/1]quit
```

- 检查 AP 状态

```
[AC_ShenZhen-wlan-view]display vap ssid wlan-net
Info: This operation may take a few seconds, please wait.
WID : WLAN ID
--------------------------------------------------------------------------------
AP ID AP name RfID WID   BSSID            Status      Auth type          STA     SSID
--------------------------------------------------------------------------------
0     area_1  0    1    AC85-3D92-3340 ON          WPA/WPA2-PSK   0       wlan-net
0     area_1  1    1    AC85-3D92-3350 ON          WPA/WPA2-PSK   1       wlan-net
1     area_2  0    1    AC85-3D92-1B60 ON          WPA/WPA2-PSK   0       wlan-net
1     area_2  1    1    AC85-3D92-1B70 ON          WPA/WPA2-PSK   0       wlan-net
```

如果是在 eNSP 中实验，则能看到模拟的信号图。笔记本电脑（STA1、STA2）搜索到名为“wlan-net”的无线网络，输入密码“Huawei@123”连接到网络。



步骤 5 配置 BGP

园区网与骨干网之间通过两条专线进行连接，并运行 BGP 路由协议。总部边界路由器 R_ShenZhen_1 和 R_ShenZhen_2 与骨干网边界路由器 R_Backbone_1 之间采用直连接口建立 EBGP 邻居，R_ShenZhen_1 和 R_ShenZhen_2 之间采用直连接口建立 IBGP 邻居。在路由器 R_ShenZhen_1 和 R_ShenZhen_2 做路由汇聚，使得路由器 R_Backbone_1 只学到 10.1.0.0/16 和 10.2.0.0/16 的聚合路由。通过 BGP 本地优先级属性控制 BGP 选路，在路由器 R_ShenZhen_1 和 R_ShenZhen_2 入向修改本地优先级属性，使得园区网去往骨干网 10.3.3.0/24 网络选择 R_ShenZhen_1 和 R_Backbone_1 之间链路，去往骨干网 10.3.4.0/24 网络选择 R_ShenZhen_2 和 R_Backbone_1 之间链路。通过 BGP MED 属性控制 BGP 选路，在路由器 R_ShenZhen_1 和 R_ShenZhen_2 出向修改 MED 属性，控制路由器 R_Backbone_1 的 BGP 选路，使得骨干网访问园区网 10.1.0.0/16 网络选择 R_ShenZhen_1 和 R_Backbone_1 之间链路，访问 DMZ 区 10.2.0.0/16 网络选择 R_ShenZhen_2 和 R_Backbone_1 之间链路。

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]ip ip-prefix 3 index 10 permit 10.3.3.0 24
[R_ShenZhen_1]route-policy 3 permit node 10
[R_ShenZhen_1-route-policy]if-match ip-prefix 3
[R_ShenZhen_1-route-policy]apply local-preference 1000
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]route-policy 3 permit node 20
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]ip ip-prefix 101 index 10 permit 10.1.0.0 16
[R_ShenZhen_1]ip ip-prefix 102 index 10 permit 10.2.0.0 16
[R_ShenZhen_1]route-policy MED permit node 10
[R_ShenZhen_1-route-policy]if-match ip-prefix 101
```

```
[R_ShenZhen_1-route-policy]apply cost 100
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]route-policy MED permit node 20
[R_ShenZhen_1-route-policy]if-match ip-prefix 102
[R_ShenZhen_1-route-policy]apply cost 200
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]bgp 65100
[R_ShenZhen_1-bgp]router-id 1.1.1.1
[R_ShenZhen_1-bgp]peer 10.1.55.2 as-number 65100
[R_ShenZhen_1-bgp]peer 172.16.1.2 as-number 65200
[R_ShenZhen_1-bgp]ipv4-family unicast
[R_ShenZhen_1-bgp-af-ipv4]aggregate 10.1.0.0 255.255.0.0 as-set detail-suppressed
[R_ShenZhen_1-bgp-af-ipv4]peer 10.1.55.2 enable
[R_ShenZhen_1-bgp-af-ipv4]peer 10.1.55.2 next-hop-local
[R_ShenZhen_1-bgp-af-ipv4]peer 172.16.1.2 enable
[R_ShenZhen_1-bgp-af-ipv4]peer 172.16.1.2 route-policy 3 import
[R_ShenZhen_1-bgp-af-ipv4]peer 172.16.1.2 route-policy MED export
```

- 配置路由器 R_ShenZhen_2

```
[R_ShenZhen_2]ip ip-prefix 4 index 10 permit 10.4.4.0 24
[R_ShenZhen_2]route-policy 4 permit node 10
[R_ShenZhen_2-route-policy]if-match ip-prefix 4
[R_ShenZhen_2-route-policy]apply local-preference 1000
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]route-policy 4 permit node 20
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]ip ip-prefix 101 index 10 permit 10.1.0.0 16
[R_ShenZhen_2]ip ip-prefix 102 index 10 permit 10.2.0.0 16
[R_ShenZhen_2]route-policy MED permit node 10
[R_ShenZhen_2-route-policy]if-match ip-prefix 101
[R_ShenZhen_2-route-policy]apply cost 200
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]route-policy MED permit node 20
[R_ShenZhen_2-route-policy]if-match ip-prefix 102
[R_ShenZhen_2-route-policy]apply cost 100
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]bgp 65100
[R_ShenZhen_2-bgp]router-id 2.2.2.2
[R_ShenZhen_2-bgp]peer 10.1.55.1 as-number 65100
[R_ShenZhen_2-bgp]peer 172.16.2.2 as-number 65200
[R_ShenZhen_2-bgp]ipv4-family unicast
[R_ShenZhen_2-bgp-af-ipv4]aggregate 10.1.0.0 255.255.0.0 as-set detail-suppressed
[R_ShenZhen_2-bgp-af-ipv4]peer 10.1.55.1 enable
[R_ShenZhen_2-bgp-af-ipv4]peer 10.1.55.1 next-hop-local
[R_ShenZhen_2-bgp-af-ipv4]peer 172.16.2.2 enable
[R_ShenZhen_2-bgp-af-ipv4]peer 172.16.2.2 route-policy 4 import
[R_ShenZhen_2-bgp-af-ipv4]peer 172.16.2.2 route-policy MED export
```

- 配置路由器 R_Backbone_1

```
[R_Backbone_1]bgp 65200
[R_Backbone_1-bgp]router-id 3.3.3.3
[R_Backbone_1-bgp]peer 172.16.1.1 as-number 65100
[R_Backbone_1-bgp]peer 172.16.2.1 as-number 65100
[R_Backbone_1-bgp]ipv4-family unicast
[R_Backbone_1-bgp-af-ipv4]network 10.3.3.0 255.255.255.0
[R_Backbone_1-bgp-af-ipv4]network 10.3.4.0 255.255.255.0
[R_Backbone_1-bgp-af-ipv4]peer 172.16.1.1 enable
[R_Backbone_1-bgp-af-ipv4]peer 172.16.2.1 enable
```

步骤 6 配置路由引入和路径控制

由于网络中存在 OSPF 和 BGP 路由协议，为了实现网络可达，需要部署路由引入，并通过路由策略进行路由优化和控制。在路由器 R_ShenZhen_1 和 R_ShenZhen_2 上均实施 BGP 和 OSPF 路由协议的双向路由引入。由于是双点双向路由引入，会带来次优路由、路由环路和路由反馈等问题，因此要实施路径控制。首先通过路由策略，对骨干网的路由在路由器 R_ShenZhen_1 和 R_ShenZhen_2 上修改优先级，只要低于 OSPF ASE 路由优先级 150 即可，本项目改为 140。为了避免路由环路，在路由引入时采用路由标记解决方案。

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]ip ip-prefix 1 index 10 permit 10.3.3.0 24
[R_ShenZhen_1]ip ip-prefix 1 index 20 permit 10.3.4.0 24
[R_ShenZhen_1]route-policy PRE permit node 10
[R_ShenZhen_1-route-policy]if-match ip-prefix 1
[R_ShenZhen_1-route-policy]apply preference 140
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]route-policy O2B deny node 10
[R_ShenZhen_1-route-policy]if-match tag 300
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]route-policy O2B permit node 20
[R_ShenZhen_1-route-policy]apply tag 400
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]route-policy B2O deny node 10
[R_ShenZhen_1-route-policy]if-match tag 100
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]route-policy B2O permit node 20
[R_ShenZhen_1-route-policy]apply tag 200
[R_ShenZhen_1-route-policy]quit
[R_ShenZhen_1]ospf 1 router-id 1.1.1.1
[R_ShenZhen_1-ospf-1]import-route bgp route-policy B2O
[R_ShenZhen_1-ospf-1]quit
[R_ShenZhen_1]bgp 65100
[R_ShenZhen_1-bgp]preference route-policy PRE
[R_ShenZhen_1-bgp]import-route ospf 1 route-policy O2B
```

- 配置路由器 R_ShenZhen_2

```
[R_ShenZhen_2]ip ip-prefix 1 index 10 permit 10.3.3.0 24
[R_ShenZhen_2]ip ip-prefix 1 index 20 permit 10.3.4.0 24
[R_ShenZhen_2]route-policy PRE permit node 10
[R_ShenZhen_2-route-policy]if-match ip-prefix 1
[R_ShenZhen_2-route-policy]apply preference 140
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]route-policy O2B deny node 10
[R_ShenZhen_2-route-policy]if-match tag 200
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]route-policy O2B permit node 20
[R_ShenZhen_2-route-policy]apply tag 100
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]route-policy B2O deny node 10
[R_ShenZhen_2-route-policy]if-match tag 200
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]route-policy B2O permit node 20
[R_ShenZhen_2-route-policy]apply tag 100
[R_ShenZhen_2-route-policy]quit
[R_ShenZhen_2]ospf 1 router-id 2.2.2.2
[R_ShenZhen_2-ospf-1]import-route bgp route-policy B2O
```

```
[R_ShenZhen_2-ospf-1]quit
[R_ShenZhen_2]bgp 65100
[R_ShenZhen_2-bgp]preference route-policy PRE
[R_ShenZhen_2-bgp]import-route ospf 1 route-policy O2B
```

步骤 7 配置 Internet 接入

为保证公司的用户连接到 Internet 不中断，园区网部署两条到 Internet 的线路，选择了不同营运商。由于不同运营商的线路质量和成本不一样，这两条线路采用主、备方式使用。园区网路由器 R_ShenZhen_1 使用主链路，路由器 R_ShenZhen_2 使用备链路。为保证主、备链路切换，采用静态路由与 NQA 联动，使用 NQA 监视主线路的状态，以便主链路异常时快速切换到备链路。

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]nqa test-instance admin icmp
[R_ShenZhen_1-nqa-admin-icmp]test-type icmp
[R_ShenZhen_1-nqa-admin-icmp]destination-address ipv4 202.96.1.1
[R_ShenZhen_1-nqa-admin-icmp]frequency 5
[R_ShenZhen_1-nqa-admin-icmp]timeout 1
[R_ShenZhen_1-nqa-admin-icmp]probe-count 2
[R_ShenZhen_1-nqa-admin-icmp]start now
[R_ShenZhen_1-nqa-admin-icmp]quit
[R_ShenZhen_1]ip route-static 0.0.0.0 0.0.0.0 202.96.1.1 track nqa admin icmp
```

- 配置路由器 R_ShenZhen_2

```
[R_ShenZhen_2]nqa test-instance admin icmp
[R_ShenZhen_2-nqa-admin-icmp]test-type icmp
[R_ShenZhen_2-nqa-admin-icmp]destination-address ipv4 202.96.1.1
[R_ShenZhen_2-nqa-admin-icmp]frequency 5
[R_ShenZhen_2-nqa-admin-icmp]interval seconds 2
[R_ShenZhen_2-nqa-admin-icmp]timeout 1
[R_ShenZhen_2-nqa-admin-icmp]probe-count 2
[R_ShenZhen_2-nqa-admin-icmp]start now
[R_ShenZhen_2-nqa-admin-icmp]quit
[R_ShenZhen_2]ip route-static 0.0.0.0 0.0.0.0 202.96.2.1 preference 100
[R_ShenZhen_2]ip route-static 0.0.0.0 0.0.0.0 10.1.55.1 track nqa admin icmp
[R_ShenZhen_2]ip route-static 202.96.1.1 255.255.255.255 10.1.55.1
```

步骤 8 配置 NAT

为了减少向 ISP 购买公网 IP 地址的成本和公司内部 IP 编址的统一规划设计，园区网和办事处内部使用的是私有 IP 地址，因此需要在园区网到 Internet 的出口路由器 R_ShenZhen_1 和 R_ShenZhen_2 以及办事处路由器上部署 NAT，使得园区网用户和办事处用户可以访问 Internet 以及外网用户访问 DMZ 区主机。园区网有两个到 Internet 的出口，重庆办事处有一个到 Internet 的出口。因为重庆办事处需要建立 IPSec VPN 访问 DMZ 区服务器的资源，所以配置使用 NAT 的 ACL 要排除掉 IPSec VPN 的感兴趣流量。公司没有申请多余的公网 IP 地址，因为边界路由器各个出口已经有固定公网 IP 地址，所以使用 Easy IP 方式部署 NAT。公司内部有 Web 服务器 10.2.100.100 需要供外网访问，园区网两个出口均要做 NAT 服务器映射。

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]acl number 3000
[R_ShenZhen_1-acl-adv-3000]rule 5 deny ip source 10.2.0.0 0.0.255.255 destination 10.4.0.0 0.0.255.255
[R_ShenZhen_1-acl-adv-3000]rule 10 permit ip source 10.0.0.0 0.255.255.255
[R_ShenZhen_1-acl-adv-3000]quit
```

```
[R_ShenZhen_1]interface GigabitEthernet1/0/0
[R_ShenZhen_1-GigabitEthernet1/0/0]nat server protocol tcp global current-interface www inside 10.2.100.100
www
[R_ShenZhen_1-GigabitEthernet1/0/0]nat outbound 3000
[R_ShenZhen_1-GigabitEthernet1/0/0]quit
```

- 配置路由器 R_ShenZhen_2

```
[R_ShenZhen_2]acl number 3000
[R_ShenZhen_2-acl-adv-3000]rule 5 deny ip source 10.2.0.0 0.0.255.255 destination 10.4.0.0 0.0.255.255
[R_ShenZhen_2-acl-adv-3000]rule 10 permit ip source 10.0.0.0 0.255.255.255
[R_ShenZhen_2-acl-adv-3000]quit
[R_ShenZhen_2]interface GigabitEthernet1/0/0
[R_ShenZhen_2-GigabitEthernet1/0/0]nat server protocol tcp global current-interface www inside 10.2.100.100
www
[R_ShenZhen_2-GigabitEthernet1/0/0]nat outbound 3000
```

- 配置路由器 R_ChongQing_1

```
[R_ChongQing_1]acl number 3000
[R_ChongQing_1-acl-adv-3000]rule 5 deny ip source 10.4.0.0 0.0.255.255 destination 10.2.0.0 0.0.255.255
[R_ChongQing_1-acl-adv-3000]rule 10 permit ip source 10.4.0.0 0.0.255.255
[R_ChongQing_1-acl-adv-3000]quit
[R_ChongQing_1]interface GigabitEthernet0/0/0
[R_ChongQing_1-GigabitEthernet0/0/0]nat outbound 3000
```

步骤 9 配置 QoS

为避免 Internet 链路视频、FTP 等流量影响公司的业务流量，需要部署 QoS 使得 Internet 链路优先保证关键应用，对 HTTP 流量和 IPSec VPN 的流量要保证带宽，HTTP 流量保证带宽为 1M，IPSec VPN 流量保证带宽为 512K。以上需求通过 MQC 的方法在两台边界路由器 R_ShenZhen_1 和 R_ShenZhen_2 上部署流量监管来实现。

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]acl number 3200
[R_ShenZhen_1-acl-adv-3200]rule 1 permit tcp source-port eq www
[R_ShenZhen_1-acl-adv-3200]quit
[R_ShenZhen_1]acl number 3201
[R_ShenZhen_1-acl-adv-3201]rule 1 permit 50
[R_ShenZhen_1-acl-adv-3201]quit
[R_ShenZhen_1]traffic classifier ipsecvpn operator or
[R_ShenZhen_1-classifier-ipsecvpn]if-match acl 3201
[R_ShenZhen_1-classifier-ipsecvpn]quit
[R_ShenZhen_1]traffic classifier http operator or
[R_ShenZhen_1-classifier-http]if-match acl 3200
[R_ShenZhen_1-classifier-http]quit
[R_ShenZhen_1]traffic behavior ipsecvpn
[R_ShenZhen_1-behavior-ipsecvpn]car cir 512 cbs 96256 pbs 160256 green pass yellow pass red discard
[R_ShenZhen_1-behavior-ipsecvpn]quit
[R_ShenZhen_1]traffic behavior http
[R_ShenZhen_1-behavior-http]car cir 1000 cbs 188000 pbs 313000 green pass yellow pass red discard
[R_ShenZhen_1-behavior-http]quit
[R_ShenZhen_1]traffic policy QOS
[R_ShenZhen_1-trafficpolicy-QOS]classifier http behavior http
[R_ShenZhen_1-trafficpolicy-QOS]classifier ipsecvpn behavior ipsecvpn
[R_ShenZhen_1-trafficpolicy-QOS]quit
[R_ShenZhen_1]interface GigabitEthernet1/0/0
[R_ShenZhen_1-GigabitEthernet1/0/0]traffic-policy QOS inbound
```

- 配置路由器 R_ShenZhen_2

```
[R_ShenZhen_2]acl number 3200
[R_ShenZhen_2-acl-adv-3200]rule 1 permit tcp source-port eq www
[R_ShenZhen_2-acl-adv-3200]quit
[R_ShenZhen_2]traffic classifier http operator or
[R_ShenZhen_2-classifier-http]if-match acl 3200
[R_ShenZhen_2-classifier-http]quit
[R_ShenZhen_2]traffic behavior http
[R_ShenZhen_2-behavior-http]car cir 1000 cbs 188000 pbs 313000 green pass yellow pass red discard
[R_ShenZhen_2-behavior-http]quit
[R_ShenZhen_2]traffic policy QOS
[R_ShenZhen_2-trafficpolicy-QOS]classifier http behavior http
[R_ShenZhen_2-trafficpolicy-QOS]quit
[R_ShenZhen_2]interface GigabitEthernet1/0/0
[R_ShenZhen_2-GigabitEthernet1/0/0]traffic-policy QOS inbound
```

步骤 10 配置 IPSec VPN

重庆办事处需要连接到 DMZ 区进行资源访问，由于专线价格的初装费和使用成本较高，因此采用 Site To Site IPSec VPN 技术作为解决方案。VPN 两端需要有固定公网 IP，会带来一定成本，但好处是 VPN 两端都可以主动建立 IPSec VPN 连接。企业总部有两个 Internet 出口，为了简化部署，本项目只在主链路连接的路由器 R_ShenZhen_1 上部署 IPSec VPN。建立 IPSec VPN 的感兴趣流量是重庆办事处的网络 10.4.0.0/16 到 DMZ 区的网络 10.2.0.0/16。

- 配置路由器 R_ShenZhen_1

```
[R_ShenZhen_1]acl number 3010
[R_ShenZhen_1-acl-adv-3010]rule 10 permit ip source 10.2.0.0 0.0.255.255 destination 10.4.0.0 0.0.255.255
[R_ShenZhen_1-acl-adv-3010]quit
[R_ShenZhen_1]ike proposal 5
[R_ShenZhen_1-ike-proposal-5]encryption-algorithm aes-cbc-128
[R_ShenZhen_1-ike-proposal-5]dh group5
[R_ShenZhen_1-ike-proposal-5]quit
[R_ShenZhen_1]ike peer RouterA v1
[R_ShenZhen_1-ike-peer-RouterA]pre-shared-key cipher 123456
[R_ShenZhen_1-ike-peer-RouterA]ike-proposal 5
[R_ShenZhen_1-ike-peer-RouterA]local-address 202.96.1.2
[R_ShenZhen_1-ike-peer-RouterA]remote-address 61.1.1.2
[R_ShenZhen_1-ike-peer-RouterA]quit
[R_ShenZhen_1]ipsec proposal tran1
[R_ShenZhen_1-ipsec-proposal-tran1]esp authentication-algorithm sha2-256
[R_ShenZhen_1-ipsec-proposal-tran1]esp encryption-algorithm aes-128
[R_ShenZhen_1-ipsec-proposal-tran1]quit
[R_ShenZhen_1]ipsec policy map1 10 isakmp
[R_ShenZhen_1-ipsec-policy-isakmp-map1-10]security acl 3010
[R_ShenZhen_1-ipsec-policy-isakmp-map1-10]ike-peer RouterA
[R_ShenZhen_1-ipsec-policy-isakmp-map1-10]proposal tran1
[R_ShenZhen_1-ipsec-policy-isakmp-map1-10]tunnel local 202.96.1.2
[R_ShenZhen_1-ipsec-policy-isakmp-map1-10]route inject dynamic
[R_ShenZhen_1-ipsec-policy-isakmp-map1-10]quit
[R_ShenZhen_1]interface GigabitEthernet1/0/0
[R_ShenZhen_1-GigabitEthernet1/0/0]ipsec policy map1
```

- 配置路由器 R_ChongQing_1

```
[R_ChongQing_1]acl number 3010
[R_ChongQing_1-acl-adv-3010]rule 10 permit ip source 10.4.0.0 0.0.255.255 destination 10.2.0.0 0.0.255.255
[R_ChongQing_1-acl-adv-3010]quit
```

```
[R_ChongQing_1]ike proposal 5
[R_ChongQing_1-ike-proposal-5]encryption-algorithm aes-cbc-128
[R_ChongQing_1-ike-proposal-5]dh group5
[R_ChongQing_1-ike-proposal-5]quit
[R_ChongQing_1]ike peer RouterB v1
[R_ChongQing_1-ike-peer-RouterB]pre-shared-key cipher 123456
[R_ChongQing_1-ike-peer-RouterB]ike-proposal 5
[R_ChongQing_1-ike-peer-RouterB]local-address 61.1.1.2
[R_ChongQing_1-ike-peer-RouterB]remote-address 202.96.1.2
[R_ChongQing_1-ike-peer-RouterB]quit
[R_ChongQing_1]ipsec proposal tran1
[R_ChongQing_1-ipsec-proposal-tran1]esp authentication-algorithm sha2-256
[R_ChongQing_1-ipsec-proposal-tran1]esp encryption-algorithm aes-128
[R_ChongQing_1-ipsec-proposal-tran1]quit
[R_ChongQing_1]ipsec policy map1 10 isakmp
[R_ChongQing_1-ipsec-policy-isakmp-map1-10]security acl 3010
[R_ChongQing_1-ipsec-policy-isakmp-map1-10]ike-peer RouterB
[R_ChongQing_1-ipsec-policy-isakmp-map1-10]proposal tran1
[R_ChongQing_1-ipsec-policy-isakmp-map1-10]tunnel local 61.1.1.2
[R_ChongQing_1-ipsec-policy-isakmp-map1-10]quit
[R_ChongQing_1]interface GigabitEthernet0/0/0
[R_ChongQing_1-GigabitEthernet0/0/0]ipsec policy map1
```

步骤 11 配置 IPv6 和 OSPFv3

为了迎接 IPv6 时代的到来，园区网络和 DMZ 区先行部署 IPv6 网络用于测试和研究，暂时不考虑连接到 Internet。仅在园区网核心交换机上和 DMZ 区的网络设备上配置路由协议实现网络互通，选择 OSPFv3 作为路由协议。本项目只关注 IPv6 路由，暂时不考虑 IPv6 的 VRRP 问题。

- 配置交换机 S_ShenZhen_1

```
[S_ShenZhen_1]ipv6
[S_ShenZhen_1]ospfv3 1
[S_ShenZhen_1-ospfv3-1]router-id 3.3.3.3
[S_ShenZhen_1-ospfv3-1]quit
[S_ShenZhen_1]interface Vlanif1
[S_ShenZhen_1-Vlanif1]ipv6 enable
[S_ShenZhen_1-Vlanif1]ipv6 address 2001:AAAA:BBBB:1::1/64
[S_ShenZhen_1-Vlanif1]ospfv3 1 area 0.0.0.0
[S_ShenZhen_1-Vlanif1]quit
[S_ShenZhen_1]interface Vlanif2
[S_ShenZhen_1-Vlanif2]ipv6 enable
[S_ShenZhen_1-Vlanif2]ipv6 address 2001:AAAA:BBBB:2::1/64
[S_ShenZhen_1-Vlanif2]ospfv3 1 area 0.0.0.0
[S_ShenZhen_1-Vlanif2]quit
[S_ShenZhen_1]interface Vlanif3
[S_ShenZhen_1-Vlanif3]ipv6 enable
[S_ShenZhen_1-Vlanif3]ipv6 address 2001:AAAA:BBBB:3::1/64
[S_ShenZhen_1-Vlanif3]ospfv3 1 area 0.0.0.0
[S_ShenZhen_1-Vlanif3]quit
[S_ShenZhen_1]interface Vlanif4
[S_ShenZhen_1-Vlanif4]ipv6 enable
[S_ShenZhen_1-Vlanif4]ipv6 address 2001:AAAA:BBBB:4::1/64
[S_ShenZhen_1-Vlanif4]ospfv3 1 area 0.0.0.0
[S_ShenZhen_1-Vlanif4]quit
[S_ShenZhen_1]interface Vlanif53
[S_ShenZhen_1-Vlanif53]ipv6 enable
```

```
[S_ShenZhen_1-Vlanif53]ipv6 address 2001:AAAA:BBBB:35::1/126
[S_ShenZhen_1-Vlanif53]ospfv3 1 area 0.0.0.0
```

- 配置交换机 S_ShenZhen_2

```
[S_ShenZhen_2]ipv6
[S_ShenZhen_2]ospfv3 1
[S_ShenZhen_2-ospfv3-1]router-id 4.4.4.4
[S_ShenZhen_2-ospfv3-1]quit
[S_ShenZhen_2]interface Vlanif1
[S_ShenZhen_2-Vlanif1]ipv6 enable
[S_ShenZhen_2-Vlanif1]ipv6 address 2001:AAAA:BBBB:1::2/64
[S_ShenZhen_2-Vlanif1]ospfv3 1 area 0.0.0.0
[S_ShenZhen_2-Vlanif1]quit
[S_ShenZhen_2]interface Vlanif2
[S_ShenZhen_2-Vlanif2]ipv6 enable
[S_ShenZhen_2-Vlanif2]ipv6 address 2001:AAAA:BBBB:2::2/64
[S_ShenZhen_2-Vlanif2]ospfv3 1 area 0.0.0.0
[S_ShenZhen_2-Vlanif2]quit
[S_ShenZhen_2]interface Vlanif3
[S_ShenZhen_2-Vlanif3]ipv6 enable
[S_ShenZhen_2-Vlanif3]ipv6 address 2001:AAAA:BBBB:3::2/64
[S_ShenZhen_2-Vlanif3]ospfv3 1 area 0.0.0.0
[S_ShenZhen_2-Vlanif3]quit
[S_ShenZhen_2]interface Vlanif4
[S_ShenZhen_2-Vlanif4]ipv6 enable
[S_ShenZhen_2-Vlanif4]ipv6 address 2001:AAAA:BBBB:4::2/64
[S_ShenZhen_2-Vlanif4]ospfv3 1 area 0.0.0.0
[S_ShenZhen_2-Vlanif4]quit
[S_ShenZhen_2]interface Vlanif54
[S_ShenZhen_2-Vlanif54]ipv6 enable
[S_ShenZhen_2-Vlanif54]ipv6 address 2001:AAAA:BBBB:36::1/126
[S_ShenZhen_2-Vlanif54]ospfv3 1 area 0.0.0.0
[S_ShenZhen_2-Vlanif54]quit
```

- 配置交换机 S_DMZ_1

```
[S_DMZ_1]ipv6
[S_DMZ_1]ospfv3 1
[S_DMZ_1-ospfv3-1]router-id 6.6.6.6
[S_DMZ_1-ospfv3-1]silent-interface Vlanif2
[S_DMZ_1-ospfv3-1]area 0.0.0.1
[S_DMZ_1-ospfv3-1-area-0.0.0.1]stub
[S_DMZ_1-ospfv3-1-area-0.0.0.1]quit
[S_DMZ_1-ospfv3-1]quit
[S_DMZ_1]interface Vlanif1
[S_DMZ_1-Vlanif1]ipv6 enable
[S_DMZ_1-Vlanif1]ipv6 address 2001:AAAA:BBBB:20::2/126
[S_DMZ_1-Vlanif1]ospfv3 1 area 0.0.0.1
[S_DMZ_1-Vlanif1]quit
[S_DMZ_1]interface Vlanif2
[S_DMZ_1-Vlanif2]ipv6 enable
[S_DMZ_1-Vlanif2]ipv6 address 2001:AAAA:BBBB:37::1/64
[S_DMZ_1-Vlanif2]ospfv3 1 area 0.0.0.1
```

- 配置交换机 S_DMZ_2

```
[S_DMZ_2]ipv6
[S_DMZ_2]ospfv3 1
[S_DMZ_2-ospfv3-1]router-id 7.7.7.7
[S_DMZ_2-ospfv3-1]silent-interface Vlanif2
```

```
[S_DMZ_2-ospfv3-1]area 0.0.0.1
[S_DMZ_2-ospfv3-1-area-0.0.0.1]stub
[S_DMZ_2-ospfv3-1-area-0.0.0.1]quit
[S_DMZ_2-ospfv3-1]quit
[S_DMZ_2]interface Vlanif1
[S_DMZ_2-Vlanif1]ipv6 enable
[S_DMZ_2-Vlanif1]ipv6 address 2001:AAAA:BBBB:21::2/126
[S_DMZ_2-Vlanif1]ospfv3 1 area 0.0.0.1
[S_DMZ_2-Vlanif1]quit
[S_DMZ_2]interface Vlanif2
[S_DMZ_2-Vlanif2]ipv6 enable
[S_DMZ_2-Vlanif2]ipv6 address 2001:AAAA:BBBB:38::1/64
[S_DMZ_2-Vlanif2]ospfv3 1 area 0.0.0.1
```

- 配置交换机 S_DMZ_3

```
[S_DMZ_3]ipv6
[S_DMZ_3]ospfv3 1
[S_DMZ_3-ospfv3-1]router-id 5.5.5.5
[S_DMZ_3-ospfv3-1]area 0.0.0.1
[S_DMZ_3-ospfv3-1-area-0.0.0.1]stub no-summary
[S_DMZ_3-ospfv3-1-area-0.0.0.1]quit
[S_DMZ_3-ospfv3-1]quit
[S_DMZ_3]interface Vlanif1
[S_DMZ_3-Vlanif1]ipv6 enable
[S_DMZ_3-Vlanif1]ipv6 address 2001:AAAA:BBBB:35::2/126
[S_DMZ_3-Vlanif1]ospfv3 1 area 0.0.0.0
[S_DMZ_3-Vlanif1]quit
[S_DMZ_3]interface Vlanif2
[S_DMZ_3-Vlanif2]ipv6 enable
[S_DMZ_3-Vlanif2]ipv6 address 2001:AAAA:BBBB:36::2/126
[S_DMZ_3-Vlanif2]ospfv3 1 area 0.0.0.0
[S_DMZ_3-Vlanif2]quit
[S_DMZ_3]interface Vlanif3
[S_DMZ_3-Vlanif3]ipv6 enable
[S_DMZ_3-Vlanif3]ipv6 address 2001:AAAA:BBBB:20::1/126
[S_DMZ_3-Vlanif3]ospfv3 1 area 0.0.0.1
[S_DMZ_3-Vlanif3]quit
[S_DMZ_3]interface Vlanif4
[S_DMZ_3-Vlanif4]ipv6 enable
[S_DMZ_3-Vlanif4]ipv6 address 2001:AAAA:BBBB:21::1/126
[S_DMZ_3-Vlanif4]ospfv3 1 area 0.0.0.1
```

12.6 项目验证

12.6.1 查看 IPv4 路由表

- 查看路由器 R_ShenZhen_1 IPv4 路由表

```
<R_ShenZhen_1>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 33        Routes : 38

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
```

```
          0.0.0.0/0   Static   60   0          RD   202.96.1.1      GigabitEthernet1/0/0
         10.1.0.0/16  IBGP     255  0          D    127.0.0.1       NULL0
         10.1.1.0/24  OSPF     10   3          D    10.1.51.1       GigabitEthernet0/0/0
                      OSPF     10   3          D    10.1.52.1       GigabitEthernet0/0/1
         10.1.2.0/24  OSPF     10   3          D    10.1.51.1       GigabitEthernet0/0/0
                      OSPF     10   3          D    10.1.52.1       GigabitEthernet0/0/1
         10.1.3.0/24  OSPF     10   3          D    10.1.51.1       GigabitEthernet0/0/0
                      OSPF     10   3          D    10.1.52.1       GigabitEthernet0/0/1
         10.1.4.0/24  OSPF     10   3          D    10.1.51.1       GigabitEthernet0/0/0
                      OSPF     10   3          D    10.1.52.1       GigabitEthernet0/0/1
        10.1.51.0/24  Direct   0    0          D    10.1.51.2       GigabitEthernet0/0/0
        10.1.51.2/32  Direct   0    0          D    127.0.0.1       GigabitEthernet0/0/0
      10.1.51.255/32  Direct   0    0          D    127.0.0.1       GigabitEthernet0/0/0
        10.1.52.0/24  Direct   0    0          D    10.1.52.2       GigabitEthernet0/0/1
        10.1.52.2/32  Direct   0    0          D    127.0.0.1       GigabitEthernet0/0/1
      10.1.52.255/32  Direct   0    0          D    127.0.0.1       GigabitEthernet0/0/1
        10.1.53.0/30  OSPF     10   3          D    10.1.51.1       GigabitEthernet0/0/0
        10.1.54.0/30  OSPF     10   3          D    10.1.52.1       GigabitEthernet0/0/1
        10.1.55.0/30  Direct   0    0          D    10.1.55.1       GigabitEthernet2/0/0
        10.1.55.1/32  Direct   0    0          D    127.0.0.1       GigabitEthernet2/0/0
        10.1.55.3/32  Direct   0    0          D    127.0.0.1       GigabitEthernet2/0/0
       10.1.100.0/24  OSPF     10   3          D    10.1.51.1       GigabitEthernet0/0/0
       10.1.200.0/24  OSPF     10   3          D    10.1.51.1       GigabitEthernet0/0/0
                      OSPF     10   3          D    10.1.52.1       GigabitEthernet0/0/1
         10.2.0.0/16  OSPF     10   4          D    10.1.52.1       GigabitEthernet0/0/1
         10.3.3.0/24  EBGP     140  0          D    172.16.1.2      GigabitEthernet0/0/2
         10.3.4.0/24  EBGP     140  0          D    172.16.1.2      GigabitEthernet0/0/2
         10.4.0.0/16  Unr      60   0          D    61.1.1.2        GigabitEthernet1/0/0
        127.0.0.0/8   Direct   0    0          D    127.0.0.1       InLoopBack0
        127.0.0.1/32  Direct   0    0          D    127.0.0.1       InLoopBack0
127.255.255.255/32    Direct   0    0          D    127.0.0.1       InLoopBack0
       172.16.1.0/30  Direct   0    0          D    172.16.1.1      GigabitEthernet0/0/2
       172.16.1.1/32  Direct   0    0          D    127.0.0.1       GigabitEthernet0/0/2
       172.16.1.3/32  Direct   0    0          D    127.0.0.1       GigabitEthernet0/0/2
       202.96.1.0/30  Direct   0    0          D    202.96.1.2      GigabitEthernet1/0/0
       202.96.1.2/32  Direct   0    0          D    127.0.0.1       GigabitEthernet1/0/0
       202.96.1.3/32  Direct   0    0          D    127.0.0.1       GigabitEthernet1/0/0
255.255.255.255/32    Direct   0    0          D    127.0.0.1       InLoopBack0
```

- 查看路由器 R_ShenZhen_2 IPv4 路由表

```
<R_ShenZhen_2>display ip routing
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 35        Routes : 39

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

          0.0.0.0/0 Static  60   0           RD  10.1.55.1       GigabitEthernet2/0/0
         10.1.0.0/16 IBGP   255  0           D   127.0.0.1       NULL0
         10.1.1.0/24 OSPF   10   3           D   10.1.51.1       GigabitEthernet0/0/0
                     OSPF   10   3           D   10.1.52.1       GigabitEthernet0/0/1
         10.1.2.0/24 OSPF   10   3           D   10.1.51.1       GigabitEthernet0/0/0
                     OSPF   10   3           D   10.1.52.1       GigabitEthernet0/0/1
         10.1.3.0/24 OSPF   10   3           D   10.1.51.1       GigabitEthernet0/0/0
                     OSPF   10   3           D   10.1.52.1       GigabitEthernet0/0/1
         10.1.4.0/24 OSPF   10   3           D   10.1.51.1       GigabitEthernet0/0/0
```

```
                    OSPF     10    3          D    10.1.52.1       GigabitEthernet0/0/1
      10.1.51.0/24  Direct   0     0          D    10.1.51.3       GigabitEthernet0/0/0
      10.1.51.3/32  Direct   0     0          D    127.0.0.1       GigabitEthernet0/0/0
    10.1.51.255/32  Direct   0     0          D    127.0.0.1       GigabitEthernet0/0/0
      10.1.52.0/24  Direct   0     0          D    10.1.52.3       GigabitEthernet0/0/1
      10.1.52.3/32  Direct   0     0          D    127.0.0.1       GigabitEthernet0/0/1
    10.1.52.255/32  Direct   0     0          D    127.0.0.1       GigabitEthernet0/0/1
      10.1.53.0/30  OSPF     10    3          D    10.1.51.1       GigabitEthernet0/0/0
      10.1.54.0/30  OSPF     10    3          D    10.1.52.1       GigabitEthernet0/0/1
      10.1.55.0/30  Direct   0     0          D    10.1.55.2       GigabitEthernet2/0/0
      10.1.55.2/32  Direct   0     0          D    127.0.0.1       GigabitEthernet2/0/0
      10.1.55.3/32  Direct   0     0          D    127.0.0.1       GigabitEthernet2/0/0
     10.1.100.0/24  OSPF     10    3          D    10.1.51.1       GigabitEthernet0/0/0
     10.1.200.0/24  OSPF     10    3          D    10.1.51.1       GigabitEthernet0/0/0
                    OSPF     10    3          D    10.1.52.1       GigabitEthernet0/0/1
       10.2.0.0/16  OSPF     10    4          D    10.1.52.1       GigabitEthernet0/0/1
       10.3.3.0/24  IBGP     140   0          RD   10.1.55.1       GigabitEthernet2/0/0
       10.3.4.0/24  EBGP     140   0          D    172.16.2.2      GigabitEthernet0/0/2
       10.4.0.0/16  O_ASE    150   1          D    10.1.55.1       GigabitEthernet2/0/0
      127.0.0.0/8   Direct   0     0          D    127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct   0     0          D    127.0.0.1       InLoopBack0
127.255.255.255/32  Direct   0     0          D    127.0.0.1       InLoopBack0
     172.16.2.0/30  Direct   0     0          D    172.16.2.1      GigabitEthernet0/0/2
     172.16.2.1/32  Direct   0     0          D    127.0.0.1       GigabitEthernet0/0/2
     172.16.2.3/32  Direct   0     0          D    127.0.0.1       GigabitEthernet0/0/2
     202.96.1.1/32  Static   60    0          RD   10.1.55.1       GigabitEthernet2/0/0
     202.96.2.0/30  Direct   0     0          D    202.96.2.2      GigabitEthernet1/0/0
     202.96.2.2/32  Direct   0     0          D    127.0.0.1       GigabitEthernet1/0/0
     202.96.2.3/32  Direct   0     0          D    127.0.0.1       GigabitEthernet1/0/0
255.255.255.255/32  Direct   0     0          D    127.0.0.1       InLoopBack0
```

- 查看交换机 S_ShenZhen_1 IPv4 路由表

```
<S_ShenZhen_1>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 29        Routes : 34

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   O_ASE   150  11           D   10.1.51.2       Vlanif51
      10.1.1.0/24   Direct  0    0            D   10.1.1.1        Vlanif1
      10.1.1.1/32   Direct  0    0            D   127.0.0.1       Vlanif1
    10.1.1.254/32   Direct  0    0            D   127.0.0.1       Vlanif1
      10.1.2.0/24   Direct  0    0            D   10.1.2.1        Vlanif2
      10.1.2.1/32   Direct  0    0            D   127.0.0.1       Vlanif2
    10.1.2.254/32   Direct  0    0            D   127.0.0.1       Vlanif2
      10.1.3.0/24   Direct  0    0            D   10.1.3.1        Vlanif3
      10.1.3.1/32   Direct  0    0            D   127.0.0.1       Vlanif3
      10.1.4.0/24   Direct  0    0            D   10.1.4.1        Vlanif4
      10.1.4.1/32   Direct  0    0            D   127.0.0.1       Vlanif4
     10.1.51.0/24   Direct  0    0            D   10.1.51.1       Vlanif51
     10.1.51.1/32   Direct  0    0            D   127.0.0.1       Vlanif51
     10.1.52.0/24   OSPF    10   3            D   10.1.51.3       Vlanif51
                    OSPF    10   3            D   10.1.51.2       Vlanif51
     10.1.53.0/30   Direct  0    0            D   10.1.53.1       Vlanif53
     10.1.53.1/32   Direct  0    0            D   127.0.0.1       Vlanif53
     10.1.54.0/30   OSPF    10   4            D   10.1.51.3       Vlanif51
```

```
                    OSPF     10    4          D    10.1.51.2       Vlanif51
      10.1.55.0/30  OSPF     10    2          D    10.1.51.3       Vlanif51
                    OSPF     10    2          D    10.1.51.2       Vlanif51
     10.1.100.0/24  Direct   0     0          D    10.1.100.1      Vlanif100
     10.1.100.1/32  Direct   0     0          D    127.0.0.1       Vlanif100
   10.1.100.254/32  Direct   0     0          D    127.0.0.1       Vlanif100
     10.1.200.0/24  Direct   0     0          D    10.1.200.1      Vlanif200
     10.1.200.1/32  Direct   0     0          D    127.0.0.1       Vlanif200
       10.2.0.0/16  OSPF     10    5          D    10.1.51.3       Vlanif51
                    OSPF     10    5          D    10.1.51.2       Vlanif51
       10.3.3.0/24  O_ASE    150   1          D    10.1.51.2       Vlanif51
       10.3.4.0/24  O_ASE    150   1          D    10.1.51.3       Vlanif51
                    O_ASE    150   1          D    10.1.51.2       Vlanif51
       10.4.0.0/16  O_ASE    150   1          D    10.1.51.2       Vlanif51
      127.0.0.0/8   Direct   0     0          D    127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct   0     0          D    127.0.0.1       InLoopBack0
```

- 查看交换机 S_ShenZhen_2 IPv4 路由表

```
<S_ShenZhen_2>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 29        Routes : 32

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   O_ASE   150  11           D   10.1.52.2       Vlanif52
       10.1.1.0/24  Direct  0    0            D   10.1.1.2        Vlanif1
       10.1.1.2/32  Direct  0    0            D   127.0.0.1       Vlanif1
       10.1.2.0/24  Direct  0    0            D   10.1.2.2        Vlanif2
       10.1.2.2/32  Direct  0    0            D   127.0.0.1       Vlanif2
       10.1.3.0/24  Direct  0    0            D   10.1.3.2        Vlanif3
       10.1.3.2/32  Direct  0    0            D   127.0.0.1       Vlanif3
     10.1.3.254/32  Direct  0    0            D   127.0.0.1       Vlanif3
       10.1.4.0/24  Direct  0    0            D   10.1.4.2        Vlanif4
       10.1.4.2/32  Direct  0    0            D   127.0.0.1       Vlanif4
     10.1.4.254/32  Direct  0    0            D   127.0.0.1       Vlanif4
      10.1.51.0/24  OSPF    10   3            D   10.1.52.3       Vlanif52
                    OSPF    10   3            D   10.1.52.2       Vlanif52
      10.1.52.0/24  Direct  0    0            D   10.1.52.1       Vlanif52
      10.1.52.1/32  Direct  0    0            D   127.0.0.1       Vlanif52
      10.1.53.0/30  OSPF    10   2            D   10.1.54.2       Vlanif54
      10.1.54.0/30  Direct  0    0            D   10.1.54.1       Vlanif54
      10.1.54.1/32  Direct  0    0            D   127.0.0.1       Vlanif54
      10.1.55.0/30  OSPF    10   2            D   10.1.52.3       Vlanif52
                    OSPF    10   2            D   10.1.52.2       Vlanif52
     10.1.100.0/24  Direct  0    0            D   10.1.100.2      Vlanif100
     10.1.100.2/32  Direct  0    0            D   127.0.0.1       Vlanif100
     10.1.200.0/24  Direct  0    0            D   10.1.200.2      Vlanif200
     10.1.200.2/32  Direct  0    0            D   127.0.0.1       Vlanif200
   10.1.200.254/32  Direct  0    0            D   127.0.0.1       Vlanif200
       10.2.0.0/16  OSPF    10   2            D   10.1.54.2       Vlanif54
       10.3.3.0/24  O_ASE   150  1            D   10.1.52.2       Vlanif52
       10.3.4.0/24  O_ASE   150  1            D   10.1.52.3       Vlanif52
                    O_ASE   150  1            D   10.1.52.2       Vlanif52
       10.4.0.0/16  O_ASE   150  1            D   10.1.52.2       Vlanif52
      127.0.0.0/8   Direct  0    0            D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0            D   127.0.0.1       InLoopBack0
```

- 查看交换机 S_DMZ_1 IPv4 路由表

```
<S_DMZ_1>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 9        Routes : 9

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   OSPF    10   2           D   10.2.1.1        Vlanif1
       10.2.1.0/30  Direct  0    0           D   10.2.1.2        Vlanif1
       10.2.1.2/32  Direct  0    0           D   127.0.0.1       Vlanif1
       10.2.2.0/30  OSPF    10   2           D   10.2.1.1        Vlanif1
       10.2.3.0/24  Direct  0    0           D   10.2.3.254      Vlanif2
     10.2.3.254/32  Direct  0    0           D   127.0.0.1       Vlanif2
       10.2.4.0/24  OSPF    10   3           D   10.2.1.1        Vlanif1
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
```

- 查看交换机 S_DMZ_2 IPv4 路由表

```
<S_DMZ_2>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 9        Routes : 9

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   OSPF    10   2           D   10.2.2.1        Vlanif1
       10.2.1.0/30  OSPF    10   2           D   10.2.2.1        Vlanif1
       10.2.2.0/30  Direct  0    0           D   10.2.2.2        Vlanif1
       10.2.2.2/32  Direct  0    0           D   127.0.0.1       Vlanif1
       10.2.3.0/24  OSPF    10   3           D   10.2.2.1        Vlanif1
       10.2.4.0/24  Direct  0    0           D   10.2.4.254      Vlanif2
     10.2.4.254/32  Direct  0    0           D   127.0.0.1       Vlanif2
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
```

- 查看交换机 S_DMZ_3 IPv4 路由表

```
<S_DMZ_3>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 23       Routes : 23

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   O_ASE   150  12          D   10.1.54.1       Vlanif2
       10.1.1.0/24  OSPF    10   2           D   10.1.54.1       Vlanif2
       10.1.2.0/24  OSPF    10   2           D   10.1.54.1       Vlanif2
       10.1.3.0/24  OSPF    10   2           D   10.1.54.1       Vlanif2
       10.1.4.0/24  OSPF    10   2           D   10.1.54.1       Vlanif2
      10.1.51.0/24  OSPF    10   4           D   10.1.54.1       Vlanif2
      10.1.52.0/24  OSPF    10   2           D   10.1.54.1       Vlanif2
      10.1.53.0/30  Direct  0    0           D   10.1.53.2       Vlanif1
      10.1.53.2/32  Direct  0    0           D   127.0.0.1       Vlanif1
      10.1.54.0/30  Direct  0    0           D   10.1.54.2       Vlanif2
```

```
      10.1.54.2/32  Direct    0    0          D   127.0.0.1       Vlanif2
      10.1.55.0/30  OSPF     10    3          D   10.1.54.1       Vlanif2
     10.1.100.0/24  OSPF     10    5          D   10.1.54.1       Vlanif2
     10.1.200.0/24  OSPF     10    2          D   10.1.54.1       Vlanif2
      10.2.1.0/30   Direct    0    0          D   10.2.1.1        Vlanif3
      10.2.1.1/32   Direct    0    0          D   127.0.0.1       Vlanif3
      10.2.2.0/30   Direct    0    0          D   10.2.2.1        Vlanif4
      10.2.2.1/32   Direct    0    0          D   127.0.0.1       Vlanif4
      10.3.3.0/24   O_ASE   150    1          D   10.1.54.1       Vlanif2
      10.3.4.0/24   O_ASE   150    1          D   10.1.54.1       Vlanif2
      10.4.0.0/16   O_ASE   150    1          D   10.1.54.1       Vlanif2
     127.0.0.0/8    Direct    0    0          D   127.0.0.1       InLoopBack0
     127.0.0.1/32   Direct    0    0          D   127.0.0.1       InLoopBack0
```

- 查看路由器 R_Backbone_1 IPv4 路由表

```
<R_Backbone_1>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 18        Routes : 18

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        10.1.0.0/16  EBGP   255   100         D   172.16.1.1      GigabitEthernet0/0/0
        10.2.0.0/16  EBGP   255   100         D   172.16.2.1      GigabitEthernet0/0/1
        10.3.3.0/24  Direct   0     0         D   10.3.3.1        LoopBack0
        10.3.3.1/32  Direct   0     0         D   127.0.0.1       LoopBack0
      10.3.3.255/32  Direct   0     0         D   127.0.0.1       LoopBack0
        10.3.4.0/24  Direct   0     0         D   10.3.4.1        LoopBack1
        10.3.4.1/32  Direct   0     0         D   127.0.0.1       LoopBack1
      10.3.4.255/32  Direct   0     0         D   127.0.0.1       LoopBack1
       127.0.0.0/8   Direct   0     0         D   127.0.0.1       InLoopBack0
       127.0.0.1/32  Direct   0     0         D   127.0.0.1       InLoopBack0
127.255.255.255/32   Direct   0     0         D   127.0.0.1       InLoopBack0
      172.16.1.0/30  Direct   0     0         D   172.16.1.2      GigabitEthernet0/0/0
      172.16.1.2/32  Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/0
      172.16.1.3/32  Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/0
      172.16.2.0/30  Direct   0     0         D   172.16.2.2      GigabitEthernet0/0/1
      172.16.2.2/32  Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/1
      172.16.2.3/32  Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/1
255.255.255.255/32   Direct   0     0         D   127.0.0.1       InLoopBack0
```

- 查看路由器 R_ChongQing_1 IPv4 路由表

```
<R_ChongQing_1>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 11        Routes : 11

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

         0.0.0.0/0   Static  60     0        RD   61.1.1.1        GigabitEthernet0/0/0
       10.4.1.0/24   Direct   0     0         D   10.4.1.254      GigabitEthernet0/0/1
     10.4.1.254/32   Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/1
     10.4.1.255/32   Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/1
       61.1.1.0/30   Direct   0     0         D   61.1.1.2        GigabitEthernet0/0/0
       61.1.1.2/32   Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/0
       61.1.1.3/32   Direct   0     0         D   127.0.0.1       GigabitEthernet0/0/0
```

```
          127.0.0.0/8   Direct  0    0          D   127.0.0.1       InLoopBack0
          127.0.0.1/32  Direct  0    0          D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0          D   127.0.0.1       InLoopBack0
255.255.255.255/32  Direct  0    0          D   127.0.0.1       InLoopBack0
```

- 查看路由器 ISP IPv4 路由表

```
<ISP>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 10        Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

          8.8.8.0/24  Direct  0    0          D   8.8.8.8         LoopBack1
          8.8.8.8/32  Direct  0    0          D   127.0.0.1       LoopBack1
         61.1.1.0/30  Direct  0    0          D   61.1.1.1        Vlanif1
         61.1.1.1/32  Direct  0    0          D   127.0.0.1       Vlanif1
        127.0.0.0/8   Direct  0    0          D   127.0.0.1       InLoopBack0
        127.0.0.1/32  Direct  0    0          D   127.0.0.1       InLoopBack0
       202.96.1.0/30  Direct  0    0          D   202.96.1.1      Vlanif2
       202.96.1.1/32  Direct  0    0          D   127.0.0.1       Vlanif2
       202.96.2.0/30  Direct  0    0          D   202.96.2.1      Vlanif3
       202.96.2.1/32  Direct  0    0          D   127.0.0.1       Vlanif3
```

12.6.2 查看 OSPFv3 路由表

- 查看交换机 S_ShenZhen_1 OSPFv3 路由表

```
[S_ShenZhen_1]display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
        N - NSSA, U - Uninstalled

OSPFv3 Process (1)
      Destination                                          Metric
        Next-hop
      2001:AAAA:BBBB:1::/64                                   1
        directly connected, Vlanif1
      2001:AAAA:BBBB:2::/64                                   1
        directly connected, Vlanif2
      2001:AAAA:BBBB:3::/64                                   1
        directly connected, Vlanif3
      2001:AAAA:BBBB:4::/64                                   1
        directly connected, Vlanif4
  IA 2001:AAAA:BBBB:20::/126                                  2
        via FE80::4E1F:CCFF:FEE3:2C82, Vlanif53
  IA 2001:AAAA:BBBB:21::/126                                  2
        via FE80::4E1F:CCFF:FEE3:2C82, Vlanif53
      2001:AAAA:BBBB:35::/126                                 1
        directly connected, Vlanif53
      2001:AAAA:BBBB:36::/126                                 2
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif4
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif3
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif2
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif1
        via FE80::4E1F:CCFF:FEE3:2C82, Vlanif53
  IA 2001:AAAA:BBBB:37::/64                                   3
```

```
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif53
   IA 2001:AAAA:BBBB:38::/64                                    3
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif53
```

- 查看交换机 S_ShenZhen_2 OSPFv3 路由表

```
[S_ShenZhen_2]display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
            N - NSSA, U - Uninstalled

OSPFv3 Process (1)
        Destination                                          Metric
          Next-hop
        2001:AAAA:BBBB:1::/64                                  1
          directly connected, Vlanif1
        2001:AAAA:BBBB:2::/64                                  1
          directly connected, Vlanif2
        2001:AAAA:BBBB:3::/64                                  1
          directly connected, Vlanif3
        2001:AAAA:BBBB:4::/64                                  1
          directly connected, Vlanif4
   IA 2001:AAAA:BBBB:20::/126                                  2
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif54
   IA 2001:AAAA:BBBB:21::/126                                  2
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif54
        2001:AAAA:BBBB:35::/126                                2
          via FE80::4E1F:CCFF:FE76:26A7, Vlanif4
          via FE80::4E1F:CCFF:FE76:26A7, Vlanif3
          via FE80::4E1F:CCFF:FE76:26A7, Vlanif2
          via FE80::4E1F:CCFF:FE76:26A7, Vlanif1
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif54
        2001:AAAA:BBBB:36::/126                                1
          directly connected, Vlanif54
   IA 2001:AAAA:BBBB:37::/64                                   3
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif54
   IA 2001:AAAA:BBBB:38::/64                                   3
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif54
```

- 查看交换机 S_DMZ_1 OSPFv3 路由表

```
[S_DMZ_1]display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
            N - NSSA, U - Uninstalled

OSPFv3 Process (1)
        Destination                                          Metric
          Next-hop
   IA ::/0                                                     2
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif1
        2001:AAAA:BBBB:20::/126                                1
          directly connected, Vlanif1
        2001:AAAA:BBBB:21::/126                                2
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif1
        2001:AAAA:BBBB:37::/64                                 1
          directly connected, Vlanif2
        2001:AAAA:BBBB:38::/64                                 3
          via FE80::4E1F:CCFF:FEE3:2C82, Vlanif1
```

- 查看交换机 S_DMZ_2 OSPFv3 路由表

```
[S_DMZ_2]display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
        N - NSSA, U - Uninstalled

OSPFv3 Process (1)
      Destination                                                    Metric
        Next-hop
  IA ::/0                                                              2
        via FE80::4E1F:CCFF:FEE3:2C82, Vlanif1
      2001:AAAA:BBBB:20::/126                                          2
        via FE80::4E1F:CCFF:FEE3:2C82, Vlanif1
      2001:AAAA:BBBB:21::/126                                          1
        directly connected, Vlanif1
      2001:AAAA:BBBB:37::/64                                           3
        via FE80::4E1F:CCFF:FEE3:2C82, Vlanif1
      2001:AAAA:BBBB:38::/64                                           1
        directly connected, Vlanif2
```

- 查看交换机 S_DMZ_3 OSPFv3 路由表

```
[S_DMZ_3]display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
        N - NSSA, U - Uninstalled

OSPFv3 Process (1)
      Destination                                                    Metric
        Next-hop
      2001:AAAA:BBBB:1::/64                                            2
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif2
        via FE80::4E1F:CCFF:FE76:26A7, Vlanif1
      2001:AAAA:BBBB:2::/64                                            2
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif2
        via FE80::4E1F:CCFF:FE76:26A7, Vlanif1
      2001:AAAA:BBBB:3::/64                                            2
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif2
        via FE80::4E1F:CCFF:FE76:26A7, Vlanif1
      2001:AAAA:BBBB:4::/64                                            2
        via FE80::4E1F:CCFF:FEC9:2903, Vlanif2
        via FE80::4E1F:CCFF:FE76:26A7, Vlanif1
      2001:AAAA:BBBB:20::/126                                          1
        directly connected, Vlanif3
      2001:AAAA:BBBB:21::/126                                          1
        directly connected, Vlanif4
      2001:AAAA:BBBB:35::/126                                          1
        directly connected, Vlanif1
      2001:AAAA:BBBB:36::/126                                          1
        directly connected, Vlanif2
      2001:AAAA:BBBB:37::/64                                           2
        via FE80::4E1F:CCFF:FE89:519E, Vlanif3
      2001:AAAA:BBBB:38::/64                                           2
        via FE80::4E1F:CCFF:FE39:151B, Vlanif4
```

12.6.3 测试连通性

- 从 PC1 测试 Internet 连通性

```
PC>ping 8.8.8.8

Ping 8.8.8.8: 32 data bytes, Press Ctrl_C to break
From 8.8.8.8: bytes=32 seq=1 ttl=253 time=374 ms
From 8.8.8.8: bytes=32 seq=2 ttl=253 time=202 ms
From 8.8.8.8: bytes=32 seq=3 ttl=253 time=608 ms
From 8.8.8.8: bytes=32 seq=4 ttl=253 time=171 ms
From 8.8.8.8: bytes=32 seq=5 ttl=253 time=733 ms

--- 8.8.8.8 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 171/417/733 ms
```

- 从 PC1 测试骨干网连通性

```
PC>ping 10.3.3.1

Ping 10.3.3.1: 32 data bytes, Press Ctrl_C to break
From 10.3.3.1: bytes=32 seq=1 ttl=253 time=718 ms
From 10.3.3.1: bytes=32 seq=2 ttl=253 time=577 ms
From 10.3.3.1: bytes=32 seq=3 ttl=253 time=406 ms
From 10.3.3.1: bytes=32 seq=4 ttl=253 time=344 ms
From 10.3.3.1: bytes=32 seq=5 ttl=253 time=47 ms

--- 10.3.3.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 47/418/718 ms

PC>ping 10.3.4.1

Ping 10.3.4.1: 32 data bytes, Press Ctrl_C to break
From 10.3.4.1: bytes=32 seq=1 ttl=253 time=234 ms
From 10.3.4.1: bytes=32 seq=2 ttl=253 time=421 ms
From 10.3.4.1: bytes=32 seq=3 ttl=253 time=109 ms
From 10.3.4.1: bytes=32 seq=4 ttl=253 time=203 ms
From 10.3.4.1: bytes=32 seq=5 ttl=253 time=172 ms

--- 10.3.4.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 109/227/421 ms
```

- 从 PC1 测试 DMZ 连通性

```
PC>ping 10.2.3.100
Ping 10.2.3.100: 32 data bytes, Press Ctrl_C to break
From 10.2.3.100: bytes=32 seq=1 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=3 ttl=252 time=109 ms
From 10.2.3.100: bytes=32 seq=4 ttl=252 time=93 ms
```

```
From 10.2.3.100: bytes=32 seq=5 ttl=252 time=78 ms

--- 10.2.3.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/87/109 ms

PC>ping 10.2.4.100

Ping 10.2.4.100: 32 data bytes, Press Ctrl_C to break
From 10.2.4.100: bytes=32 seq=1 ttl=252 time=94 ms
From 10.2.4.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=3 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=4 ttl=252 time=109 ms
From 10.2.4.100: bytes=32 seq=5 ttl=252 time=94 ms

--- 10.2.4.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/90/109 ms
```

- 从 PC2 测试 Internet 连通性

```
PC>ping 8.8.8.8

Ping 8.8.8.8: 32 data bytes, Press Ctrl_C to break
From 8.8.8.8: bytes=32 seq=1 ttl=253 time=47 ms
From 8.8.8.8: bytes=32 seq=2 ttl=253 time=78 ms
From 8.8.8.8: bytes=32 seq=3 ttl=253 time=78 ms
From 8.8.8.8: bytes=32 seq=4 ttl=253 time=94 ms
From 8.8.8.8: bytes=32 seq=5 ttl=253 time=94 ms

--- 8.8.8.8 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 47/78/94 ms
```

- 从 PC2 测试骨干网连通性

```
PC>ping 10.3.3.1

Ping 10.3.3.1: 32 data bytes, Press Ctrl_C to break
From 10.3.3.1: bytes=32 seq=1 ttl=253 time=62 ms
From 10.3.3.1: bytes=32 seq=2 ttl=253 time=62 ms
From 10.3.3.1: bytes=32 seq=3 ttl=253 time=63 ms
From 10.3.3.1: bytes=32 seq=4 ttl=253 time=78 ms
From 10.3.3.1: bytes=32 seq=5 ttl=253 time=47 ms

--- 10.3.3.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 47/62/78 ms

PC>ping 10.3.4.1

Ping 10.3.4.1: 32 data bytes, Press Ctrl_C to break
```

```
From 10.3.4.1: bytes=32 seq=1 ttl=253 time=47 ms
From 10.3.4.1: bytes=32 seq=2 ttl=253 time=62 ms
From 10.3.4.1: bytes=32 seq=3 ttl=253 time=47 ms
From 10.3.4.1: bytes=32 seq=4 ttl=253 time=78 ms
From 10.3.4.1: bytes=32 seq=5 ttl=253 time=47 ms

--- 10.3.4.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 47/56/78 ms
```

- 从 PC2 测试 DMZ 连通性

```
PC>ping 10.2.3.100

Ping 10.2.3.100: 32 data bytes, Press Ctrl_C to break
From 10.2.3.100: bytes=32 seq=1 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=3 ttl=252 time=109 ms
From 10.2.3.100: bytes=32 seq=4 ttl=252 time=93 ms
From 10.2.3.100: bytes=32 seq=5 ttl=252 time=78 ms

--- 10.2.3.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/87/109 ms

PC>ping 10.2.4.100

Ping 10.2.4.100: 32 data bytes, Press Ctrl_C to break
From 10.2.4.100: bytes=32 seq=1 ttl=252 time=94 ms
From 10.2.4.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=3 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=4 ttl=252 time=109 ms
From 10.2.4.100: bytes=32 seq=5 ttl=252 time=94 ms

--- 10.2.4.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/90/109 ms
```

- 从 PC3 测试 Internet 连通性

```
PC>ping 8.8.8.8

Ping 8.8.8.8: 32 data bytes, Press Ctrl_C to break
From 8.8.8.8: bytes=32 seq=1 ttl=253 time=94 ms
From 8.8.8.8: bytes=32 seq=2 ttl=253 time=63 ms
From 8.8.8.8: bytes=32 seq=3 ttl=253 time=63 ms
From 8.8.8.8: bytes=32 seq=4 ttl=253 time=47 ms
From 8.8.8.8: bytes=32 seq=5 ttl=253 time=47 ms

--- 8.8.8.8 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
```

```
   round-trip min/avg/max = 47/62/94 ms
```

- 从 PC3 测试骨干网连通性

```
PC>ping 10.3.3.1
Ping 10.3.3.1: 32 data bytes, Press Ctrl_C to break
From 10.3.3.1: bytes=32 seq=1 ttl=253 time=62 ms
From 10.3.3.1: bytes=32 seq=2 ttl=253 time=47 ms
From 10.3.3.1: bytes=32 seq=3 ttl=253 time=47 ms
From 10.3.3.1: bytes=32 seq=4 ttl=253 time=63 ms
From 10.3.3.1: bytes=32 seq=5 ttl=253 time=62 ms

--- 10.3.3.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 47/56/63 ms

PC>ping 10.3.4.1

Ping 10.3.4.1: 32 data bytes, Press Ctrl_C to break
From 10.3.4.1: bytes=32 seq=1 ttl=253 time=47 ms
From 10.3.4.1: bytes=32 seq=2 ttl=253 time=46 ms
From 10.3.4.1: bytes=32 seq=3 ttl=253 time=78 ms
From 10.3.4.1: bytes=32 seq=4 ttl=253 time=47 ms
From 10.3.4.1: bytes=32 seq=5 ttl=253 time=78 ms

--- 10.3.4.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 46/59/78 ms
```

- 从 PC3 测试 DMZ 连通性

```
PC>ping 10.2.3.100
Ping 10.2.3.100: 32 data bytes, Press Ctrl_C to break
From 10.2.3.100: bytes=32 seq=1 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=3 ttl=252 time=109 ms
From 10.2.3.100: bytes=32 seq=4 ttl=252 time=93 ms
From 10.2.3.100: bytes=32 seq=5 ttl=252 time=78 ms

--- 10.2.3.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/87/109 ms

PC>ping 10.2.4.100

Ping 10.2.4.100: 32 data bytes, Press Ctrl_C to break
From 10.2.4.100: bytes=32 seq=1 ttl=252 time=94 ms
From 10.2.4.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=3 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=4 ttl=252 time=109 ms
From 10.2.4.100: bytes=32 seq=5 ttl=252 time=94 ms

--- 10.2.4.100 ping statistics ---
   5 packet(s) transmitted
```

```
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/90/109 ms
```

- 从 PC4 测试 Internet 连通性

```
PC>ping 8.8.8.8

Ping 8.8.8.8: 32 data bytes, Press Ctrl_C to break
From 8.8.8.8: bytes=32 seq=1 ttl=253 time=78 ms
From 8.8.8.8: bytes=32 seq=2 ttl=253 time=78 ms
From 8.8.8.8: bytes=32 seq=3 ttl=253 time=78 ms
From 8.8.8.8: bytes=32 seq=4 ttl=253 time=47 ms
From 8.8.8.8: bytes=32 seq=5 ttl=253 time=62 ms

--- 8.8.8.8 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 47/68/78 ms
```

- 从 PC4 测试骨干网连通性

```
PC>ping 10.3.3.1

Ping 10.3.3.1: 32 data bytes, Press Ctrl_C to break
From 10.3.3.1: bytes=32 seq=1 ttl=253 time=63 ms
From 10.3.3.1: bytes=32 seq=2 ttl=253 time=78 ms
From 10.3.3.1: bytes=32 seq=3 ttl=253 time=78 ms
From 10.3.3.1: bytes=32 seq=4 ttl=253 time=31 ms
From 10.3.3.1: bytes=32 seq=5 ttl=253 time=62 ms

--- 10.3.3.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 31/62/78 ms

PC>ping 10.3.4.1

Ping 10.3.4.1: 32 data bytes, Press Ctrl_C to break
From 10.3.4.1: bytes=32 seq=1 ttl=253 time=93 ms
From 10.3.4.1: bytes=32 seq=2 ttl=253 time=62 ms
From 10.3.4.1: bytes=32 seq=3 ttl=253 time=78 ms
From 10.3.4.1: bytes=32 seq=4 ttl=253 time=93 ms
From 10.3.4.1: bytes=32 seq=5 ttl=253 time=47 ms

--- 10.3.4.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 47/74/93 ms
```

- 从 PC4 测试 DMZ 连通性

```
PC>ping 10.2.3.100
Ping 10.2.3.100: 32 data bytes, Press Ctrl_C to break
From 10.2.3.100: bytes=32 seq=1 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.3.100: bytes=32 seq=3 ttl=252 time=109 ms
```

```
From 10.2.3.100: bytes=32 seq=4 ttl=252 time=93 ms
From 10.2.3.100: bytes=32 seq=5 ttl=252 time=78 ms

--- 10.2.3.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/87/109 ms

PC>ping 10.2.4.100

Ping 10.2.4.100: 32 data bytes, Press Ctrl_C to break
From 10.2.4.100: bytes=32 seq=1 ttl=252 time=94 ms
From 10.2.4.100: bytes=32 seq=2 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=3 ttl=252 time=78 ms
From 10.2.4.100: bytes=32 seq=4 ttl=252 time=109 ms
From 10.2.4.100: bytes=32 seq=5 ttl=252 time=94 ms

--- 10.2.4.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 78/90/109 ms
```

- 从 PC5 测试 Internet 连通性

```
PC>ping 8.8.8.8

Ping 8.8.8.8: 32 data bytes, Press Ctrl_C to break
From 8.8.8.8: bytes=32 seq=1 ttl=254 time=31 ms
From 8.8.8.8: bytes=32 seq=2 ttl=254 time=63 ms
From 8.8.8.8: bytes=32 seq=3 ttl=254 time=47 ms
From 8.8.8.8: bytes=32 seq=4 ttl=254 time=47 ms
From 8.8.8.8: bytes=32 seq=5 ttl=254 time=32 ms

--- 8.8.8.8 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 31/44/63 ms
```

- 从 PC5 测试 DMZ 连通性

```
PC>ping 10.2.3.100

Ping 10.2.3.100: 32 data bytes, Press Ctrl_C to break
From 10.2.3.100: bytes=32 seq=1 ttl=251 time=125 ms
From 10.2.3.100: bytes=32 seq=2 ttl=251 time=110 ms
From 10.2.3.100: bytes=32 seq=3 ttl=251 time=109 ms
From 10.2.3.100: bytes=32 seq=4 ttl=251 time=125 ms
From 10.2.3.100: bytes=32 seq=5 ttl=251 time=156 ms

--- 10.2.3.100 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 109/125/156 ms

PC>ping 10.2.4.100
```

```
Ping 10.2.4.100: 32 data bytes, Press Ctrl_C to break
From 10.2.4.100: bytes=32 seq=1 ttl=251 time=327 ms
From 10.2.4.100: bytes=32 seq=2 ttl=251 time=124 ms
From 10.2.4.100: bytes=32 seq=3 ttl=251 time=140 ms
From 10.2.4.100: bytes=32 seq=4 ttl=251 time=124 ms
From 10.2.4.100: bytes=32 seq=5 ttl=251 time=156 ms

--- 10.2.4.100 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 124/174/327 ms
```

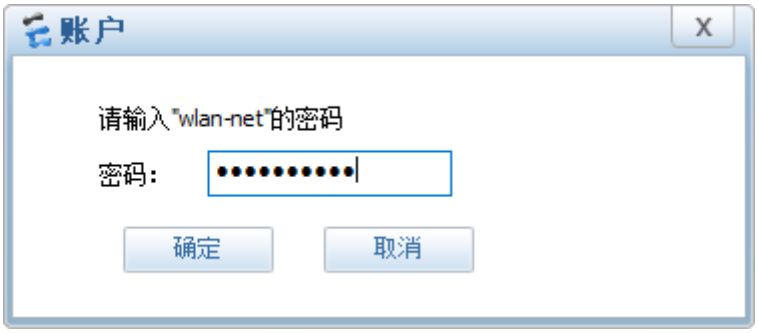
- 从骨干网测试 DMZ 连通性

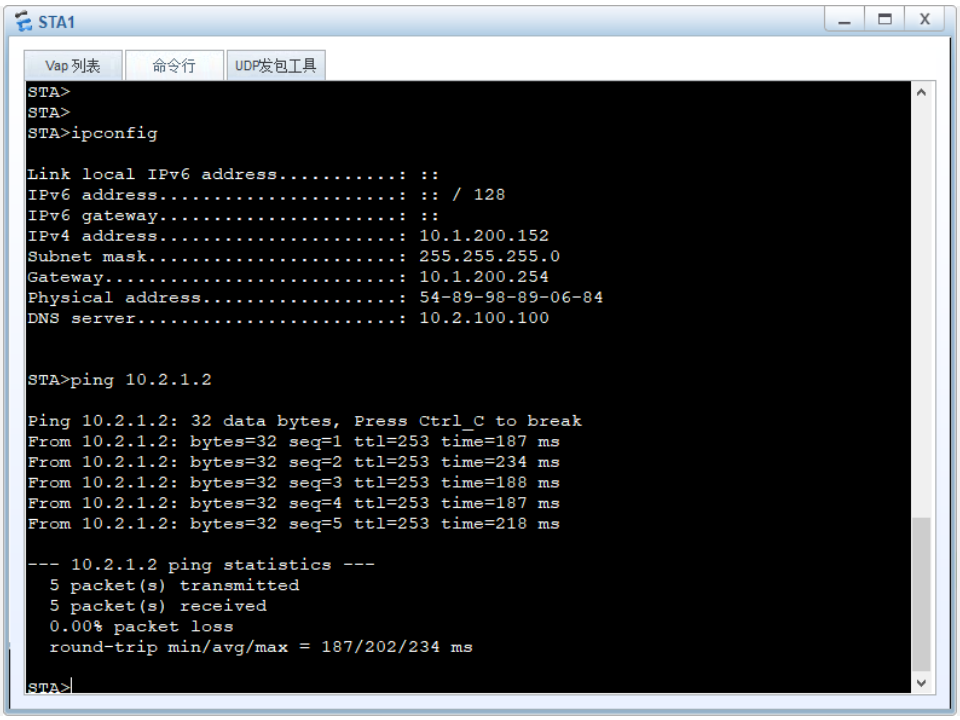
```
<R_Backbone_1>ping -a 10.3.3.1 10.2.3.100
  PING 10.2.3.100: 56   data bytes, press CTRL_C to break
    Reply from 10.2.3.100: bytes=56 Sequence=1 ttl=251 time=70 ms
    Reply from 10.2.3.100: bytes=56 Sequence=2 ttl=251 time=120 ms
    Reply from 10.2.3.100: bytes=56 Sequence=3 ttl=251 time=80 ms
    Reply from 10.2.3.100: bytes=56 Sequence=4 ttl=251 time=100 ms
    Reply from 10.2.3.100: bytes=56 Sequence=5 ttl=251 time=80 ms

  --- 10.2.3.100 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 70/90/120 ms

<R_Backbone_1>ping -a 10.3.4.1 10.2.3.100
  PING 10.2.3.100: 56   data bytes, press CTRL_C to break
    Reply from 10.2.3.100: bytes=56 Sequence=1 ttl=251 time=100 ms
    Reply from 10.2.3.100: bytes=56 Sequence=2 ttl=251 time=80 ms
    Reply from 10.2.3.100: bytes=56 Sequence=3 ttl=251 time=60 ms
    Reply from 10.2.3.100: bytes=56 Sequence=4 ttl=251 time=90 ms
    Reply from 10.2.3.100: bytes=56 Sequence=5 ttl=251 time=110 ms

  --- 10.2.3.100 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 60/88/110 ms
```

- 从 STA1、STA2 测试网络连通性

  在 STA1、STA2 上，可以看到 Vap 列表，双击 SSID，输入密码：Huawei@123，“确定”。

STA1
Vap 列表
命令行
UDP发包工具
MAC 地址:
54-89-98-89-06-84
IPv4 配置
静态
DHCP
IP 地址:
子网掩码:
网关:
Vap列表
SSID
加密方式
状态
VAP MAC
信道
射频类型
wlan-net
NULL
未连接
AC-85-3D-92-33-40
6
802.11bgn
wlan-net
NULL
未连接
AC-85-3D-92-33-50
149
连接
断开
更新
应用


账户
请输入"wlan-net"的密码
密码:
确定
取消


STA1
Vap 列表
命令行
UDP发包工具
STA>
STA>
STA>ipconfig
Link local IPv6 address.....: ::
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.1.200.152
Subnet mask.....: 255.255.255.0
Gateway.....: 10.1.200.254
Physical address.....: 54-89-98-89-06-84
DNS server.....: 10.2.100.100
STA>ping 10.2.1.2
Ping 10.2.1.2: 32 data bytes, Press Ctrl_C to break
From 10.2.1.2: bytes=32 seq=1 ttl=253 time=187 ms
From 10.2.1.2: bytes=32 seq=2 ttl=253 time=234 ms
From 10.2.1.2: bytes=32 seq=3 ttl=253 time=188 ms
From 10.2.1.2: bytes=32 seq=4 ttl=253 time=187 ms
From 10.2.1.2: bytes=32 seq=5 ttl=253 time=218 ms
--- 10.2.1.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 187/202/234 ms
STA>